Agentic Recommender Systems: A Systematic Literature Review

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***Abstract*—Recommender systems (RSs) are software systems that use machine learning techniques to suggest items, such as movies, products or routes to users based on input provided over time. These systems have been widely used by companies, such as Amazon, Google, and Netflix, who rely heavily on recommendations as a core element of their interaction with users. In the case of software engineering, RSs are used to recommend, for example, software tasks, developers, APIs, libraries, and bug fixes. More recently, advancements in natural language processing and large language models (LLMs) have led to the creation of LLM agents capable of generating plans, interacting with users and the environment, receiving feedback, and using tools. These developments have resulted in the use of LLM agents in RSs, giving rise to multi-agent RSs referred to as agentic recommender systems (ARSs). However, agentic RSs are not yet well characterized in terms of the basic elements of system modeling and design. This systematic literature review characterizes agentic recommender systems with respect to agents, roles, relationships, prompts, integration, use cases, strategies and evaluation methods. An analysis of published studies yields insights, such as (i) the identification of 13 distinct types of agents, (ii) the popularity of GPT models, (iii) a framework for ARSs, (iv) a generalized prompt, (v) the use of persona, cues, output format, and zero-shot prompting, (vi) the frequent use of Amazon, Yelp, and MovieLens datasets, (vii) the use of nDCG@k and Recall@k as common evaluation metrics, and (viii) the identification of many research gaps. This systematic literature review aims to support the continued research and development of agentic recommender systems.**

***Index Terms*—Agentic AI, LLM agent, multi-agent, LLM, large language model, recommender system, systematic literature review.**

I. Introduction

Agentic Recommender Systems (ARSs) are a novel research field that relates to the use of large language model (LLM) agents in recommender systems (RSs) to improve performance, enhance system structure and mitigate process complexity. From a software engineering point of view, ARSs are built upon traditional RSs, which are widely used in industry and academia to recommend, for example, tasks, developers, APIs, libraries, and bug fixes. In an agentic RS, LLM agents collaborate, and sometimes compete, to provide recommendations based on user and item data, drawing on internal and external knowledge sources. The development and evaluation of these systems remain an active area of research, with the promise of yielding many useful results.

However, the field of ARSs is still in its early stages and, as such, has not been thoroughly characterized. ChatGPT[1], based on the GPT models, became available to the general public in 2022 and many other models have subsequently been released. AutoGPT[2], one of the first LLM agents, also based on the GPT model, was released in early 2023. Owing to the novelty of the fields of LLMs, LLM agents, and ARSs and the rapidly evolving technologies that support them, there is limited understanding of the agents used in ARSs, their recommendation processes, and their evaluation methods. For example, several open research questions arise from the existing literature:

1) What LLM agents are used in ARSs? How are they characterized? [1], [2]
2) What prompts are used for LLM agents in ARSs? What structures do they typically follow? [3]
3) Is there variability in how LLM agents integrate with tools, algorithms, APIs, or other systems? [4], [5]
4) What metrics are used to evaluate ARSs? Are there identifiable trends? [6], [7]

This systematic literature review seeks to address these questions by characterizing agentic recommender systems reported in the literature and analyzing their implementation from different perspectives, including agents, prompts, integrations, and evaluation. It also aims to identify trends that may guide both early-career and experienced researchers and practitioners in this emerging field.

An agentic recommender system is a multi-agent system that provides recommendations based on a set of LLM agents, that is, autonomous agents that use LLMs to perform reasoning, decision making, and action execution [8]. These agents are typically defined in terms of a few characteristics, namely an LLM, prompts, input data, memory, and tools. Note that, in this review, we include agentic RSs that involve a single LLM agent. The scope of this systematic review excludes studies

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[1]https://chatgpt.com
[2]https://github.com/Significant-Gravitas/AutoGPT

based on the traditional definitions of AI agents [9] or multi-agent systems (MAS) [10].

Our contributions include:

- a characterization of agentic recommender systems reported in the literature.
- the identification of agents, agent relationships, and LLMs used in ARSs.
- an analysis of ARS usage trends from different perspectives, including recommendation strategies, agent prompts and system integrations.
- an overview of popular metrics used in ARSs, including their values, baselines and associated repositories.
- a summary of existing literature reviews on ARSs.
- a description and analysis of research gaps, open research questions, and research directions for future work.

This paper is organized as follows: Section II discusses key concepts related to ARSs, including RSs, Agentic AI, and ARSs themselves. Section III presents the methodology, protocol, and results of the systematic literature review, along with a discussion of the findings, including research gaps. Section IV outlines threats to validity. Finally, Section V concludes and outlines directions for future work.

II. Related Work

A. Recommender Systems

Recommender systems (RSs) use machine learning (ML) techniques to provide personalized item recommendations to users based on factors such as user preferences, item information, user similarities, item similarity, or user-item interactions [11], [12]. RSs are widely applied in everyday contexts, with applications across domains such as movies [13], products [14], social networks [15], and multi-domain settings [16]. Traditionally, RS algorithms are categorized into three main strategies: content-based filtering [17], which primarily relies on item features; collaborative filtering [18], which is based on similarities in user preferences; and hybrid approaches [19], which combine elements of both.

B. Agentic AI

Large Language Models (LLMs) are deep learning models trained on vast amounts of textual data to generate human-like language with high fidelity [20]. Advancements in the field of LLMs have led to the emergence of LLM agents and the advent of agentic artificial intelligence (AI). In agentic AI, LLM agents, which we call agents for short in this review, leverage LLMs to perform reasoning, decision-making, and action execution [8]. These agents integrate planning, memory management, and tool usage, and are designed to execute multi-step tasks, retrieve information, interact with external APIs or knowledge bases, and collaborate [8]. They are used in various domains, including software engineering [21], science [22], and finance [23].

C. Agentic Recommender Systems

Agentic Recommender Systems (ARSs) leverage collaboration between LLM agents to suggest relevant items to users based on users' historical interactions, item information, user information, or other contextual information, such as other systems or the internet [24]. This novel type of recommender system emerges from the collaborative application of LLM agents within the recommender system domain. Although one LLM agent with its planning and acting capabilities, access to memory, APIs, and advanced language and reasoning abilities, is enough to characterize an ARS, the main challenge, promising applications, and future directions of this research field lie in the collaboration of multiple LLM agents, their choice, placement, and recommendation workflow to provide accurate and effective recommendations to users.

The plethora of arrangements makes it challenging to provide a precise definition of an ARS, but there is a common conceptual outline [25] that is usually followed. Formally, an LLM agent can be defined in terms of its choice of LLM model m, prompt p, memory M, input data I, and a tool list T. The prompt p is defined based on the goal g, an example list E, and an action list $A = a_1, \ldots, a_{n_a}$. The LLM agent is then a function that takes these parameters and outputs either a single recommendation, a list of recommendations, or a message to a subsequent LLM agent within the recommendation workflow. The results are summarized as an output o and an action $a \in A$, which this LLM agent then performs.

$$\text{LLM Agent} : f(m, p(g, E, A), M, I, T) \rightarrow \{o, a\} \tag{1}$$

A generic agentic system can then be defined as a set of LLM agents $\mathcal{A}$ and the relationships among these agents $\mathcal{R} \subseteq \mathcal{A} \times \mathcal{A}$, where agents collaborate to achieve a high-level goal G.

$$\text{Agentic System} : f(\mathcal{A}, \mathcal{R}, G) \tag{2}$$

An agentic recommender system (ARS) is an agentic system whose goal G is to provide accurate and effective recommendations.

This definition has several points of discussion that remain unaddressed to reflect the variability in the implementation of an agentic system and, by extension, an ARS. We now elaborate on this variability:

a) A single LLM agent is an agentic system: It is worth considering whether a single LLM agent can constitute an agentic system. Collaboration is limited, even though it may be an essential feature for handling complex tasks. This study accepts ARSs defined with a single LLM agent.

b) The agents of an agentic system may have different LLMs: There are no restrictions on the types or versions of LLMs employed by agents. In an agentic system, all agents may have the same version of an LLM or they may differ to enable specialized behavior.

c) Memory is optional: Memory can be local to an LLM agent as well as global (i.e., system-wide), or entirely absent. LLMs themselves contain sufficient knowledge to answer many questions and possess advanced reasoning capabilities that can be leveraged. However, not including memory at all reduces personalization.

d) Prompt structure is flexible: A prompt is the main way to communicate with LLM agents, guide their reasoning, and limit their action scope. The concept of a well-defined prompt remains under active investigation, with numerous studies exploring optimal structures, elements, and patterns to enhance performance. This has given rise to the field of prompt engineering [26]. Commonly considered elements include the agent's goal (g), a set of examples (E), and an action space (A). While these components are optional, incorporating them often improves prompt clarity and can significantly enhance agent performance.

e) There are different ways to provide input data: Input data for an LLM agent can be specified via an API, embedded in the prompt, or incorporated through fine-tuning the LLM used in the agent. In some specialized scenarios, the LLM can be trained from scratch so that the input data becomes part of its knowledge base. Although this is rarely feasible, research in the area of small language models (SLM) is investigating solutions [27], [28].

f) Tools can be systems: LLM agents can query an algorithm provided in the tool list for some efficient or precise mathematical results, but they can also interact with external systems via suitable APIs. Both can be defined in the tool list.

Agentic RSs do not follow traditional recommendation strategies. When recommending, in general, a user-item interaction matrix is queried, as well as user and item information, by a recommendation algorithm. This algorithm may calculate a user preference and compare it with other users or items, alongside user or item information, to predict items that may be relevant for the user, which typically falls into collaborative filtering, content-based filtering, or hybrid approaches. In agentic RSs, agents may collaborate to construct the user-item interaction matrix, build preference profiles, or gather user and item information, which a specialized LLM agent then processes to generate a recommendation list. Unless this specialized LLM agent uses traditional algorithms or a traditional recommender system, it remains difficult to explain how recommendation lists are generated based on the data provided. For this reason, it is not possible to identify recommendation strategies used or classify them. Instead, a new LLM-based strategy, based on either user or item information, or other types of information can be reported.

Agentic RSs are a relatively novel research field and, for this reason, little is known about the system components that are commonly involved in its development. At a minimum, an LLM agent is used. Other prospective components include a knowledge base to store recommendation data, a coordinator of activities (which may be a framework for agentic AI or another LLM agent), and a user interface. The lack of information around such a novel and promising field provides the motivation for this study.

Relevant ARSs published in the literature are discussed in this systematic literature review. In the next section, we provide existing reviews of the literature, whether systematic or not, and situate our systematic review in relation to them.

D. Existing Reviews of the Literature on Agentic Recommender Systems

The use of LLMs in recommendation tasks is thoroughly investigated in [29]. Specific aspects, such as LLM architectures, are examined in [30]. Security, a growing area of LLM research, is reviewed in [31]. The survey in [32] discusses several methods for information retrieval that can be extended to the recommendation domain. Personalization and its open challenges are examined in [33], and [34] presents several applications of LLMs in recommendation research.

The evolution from traditional to LLM-based recommendations before the emergence of the first agentic RSs is also the subject of reviews. The authors of [35] review the recommendation workflow incorporating LLMs from the perspective of recommender systems. The evolution of recommendation algorithms is discussed in [36] and data manipulation in novel systems is reviewed in [37]. Finally, applications of chatbot-based recommender systems are discussed in detail in [38].

Novel LLM agents in recommendation tasks are also the subject of several studies. The roles and functions of LLM agents is reviewed in [39], while several applications are examined in [40].

However, there is a lack of a literature review addressing agentic recommender systems, which provide recommendations based on a set of LLM agents, that is, autonomous agents that use LLMs to perform reasoning, decision making, and action execution [8].

This systematic review of the literature differs from other surveys and reviews in its perspective and scope. First, although some studies review the literature on the use of LLM agents in recommender systems from the perspective of the agents, this study adopts the perspective of the system, an agentic RS. This shift in perspective enables discussion of technical issues, such as system integrations, use cases, and evaluation studies. Second, we adopt a broad scope in the study and analyze multiple aspects of agentic RSs, namely agent roles and relationships, prompts, integrations, use cases, recommendation strategies, and evaluation studies. Finally, we identify research gaps, trends, open research questions, and directions for future research.

III. A Systematic Review of Agentic Recommender Systems

In this section, we describe the systematic literature review in detail. The section is divided into three subsections: Method, Demographics, and Results. The first subsection addresses research questions and issues related to the study protocol. The second subsection presents the studies and analyzes demographic information. The third subsection discusses the results of each research question providing insights. Throughout, we simplify the terminology by referring to LLM agents simply as agents.

A. Method

This section details the study protocol. It outlines the research questions, document sources, search strategy, inclusion and

TABLE I
THE RESEARCH QUESTIONS

RQ		How to characterize agentic recommender systems published in the literature from a researcher's perspective with respect to their agents, agent relationships, prompts, integrations, use cases, recommendation strategies, evaluation studies, and research gaps in order to support recommender system research and development.
Agents	RQ 1	What agents are used in agentic recommender systems?
	RQ 2	What roles do agents used in agentic recommender systems play?
	RQ 3	What large language models (LLM) are used in agentic recommender systems?
	RQ 4	What knowledge augmentation strategies are used in agents used in agentic recommender systems?
Agent relationships	RQ 5	How to characterize agent relationships in agentic recommender systems?
Prompts	RQ 6	How are prompts characterized with respect to their structure, such as persona, instruction, input data, context, steps, cues, output format, and the presence of examples?
	RQ 7	What are the main prompt engineering strategies followed in agents?
Integrations	RQ 8	What integrations are agentic recommender systems reported to have?
Use cases	RQ 9	What use cases are agentic recommender systems reported to have?
Recommendation	RQ 10	What recommendation strategies are agentic recommender systems reported to have?
Evaluation	RQ 11	What datasets are used in agentic recommender system evaluation and in what domains are agentic recommender systems evaluated?
	RQ 12	What metrics are used in agentic recommender systems evaluation?
	RQ 13	What baselines are used in agentic recommender systems evaluation?
	RQ 14	What nDCG and Recall values for agentic recommender system evaluation are reported?
Research gaps reported	RQ 15	What research gaps are reported in the studies?

exclusion criteria, and data extracted from the reviewed studies. Here, we follow the guidelines for systematic literature reviews provided in [41].

As a high-level guide, we follow the Goal-Question-Metric (GQM) approach [42]. Our goal (G) is discussed next, while the questions (Q) correspond to the research questions. Lastly, metrics (M) are detailed in a subsection on the data extracted from studies.

1) Goal: Essentially, our goal is to investigate the development of agentic frameworks and characterize them in relevant ways. To structure our investigation, we use a more structured approach, following a strategy based on the PICO or PICOC framework [43], [44]:

Population: Agentic recommender systems
Investigation: Characterization
Context: Studies published in the literature
Outcome: Agents, agent relationships, prompts, integrations, use cases, recommendation strategies, evaluation studies, and research gaps.

The strategy omits two additional fields commonly included in published studies: purpose and viewpoint. These are:

Purpose: To assist in research and development of agentic recommender systems
Viewpoint: Researcher

These elements are combined to define our goal, which is used as the main research question. This is presented in the next section together with other specific research questions.

2) Research Questions: In this section, we detail all research questions that guide this review. As presented in the previous section, our main goal is to characterize agentic recommender systems from multiple perspectives. These include the agents used in ARSs, agent relationships, prompts, system integrations with other platforms, use cases in which ARSs assist users, recommendation strategies, evaluation studies, and research gaps reported in the literature. We then develop several research questions for each perspective and present them in Table I.

The table begins with the main research question, derived from our goal. Then, it lists the research questions for each investigation perspective. One of the first objectives is to identify agents used in ARSs and how they maintain relationships, which correspond to the first and second perspectives. Research questions 1-4 identify LLM agents and characterize them with respect to their roles, underlying LLMs, and applied knowledge augmentation techniques. Agent relationships are characterized by analyzing common connections among agents.

Prompts used in agentic RSs are also identified. They are characterized based on their format, elements such as persona, context, output format, and strategies. Next, we identify the integrations of agentic RSs and their usage. Integrations are related to connections to external systems, databases, or algorithms. Use cases are the types of recommendations, or other actions, that motivate users when using a recommender system, such as seeking recommendation or providing feedback.

The recommendations themselves are characterized with respect to their traditional strategies, such as collaborative filtering-based, or novel LLM-based strategies. We anticipate that the use of LLMs renders traditional strategies difficult to distinguish, thus favoring LLM-based recommendation strategies. The evaluation of ARSs is characterized based on datasets, domains, metrics, and baselines. Finally, we present research gaps identified in the reviewed studies.

3) Document Sources: Four scientific search engines were selected for their popularity and efficiency for computer science research. These include Scopus[3], Web of Science[4], ACM Digital Library[5], and IEEE Xplore[6].

Scopus provides a feature that searches for scientific preprints on arXiv[7] and similar repositories. We use this feature and include the returned preprints in this systematic review to maximize coverage of available studies. However, we acknowledge that some preprints are not peer-reviewed, which may affect quality. arXiv preprints are explicitly indicated in the next subsection. Google Scholar[8] is not included in this study because of its lack of control over the search string.

4) Search Strategy: In this review, the search strategy has five steps. The first is to specify the search string. The search string is applied to publication titles, abstracts, and keywords. It was constructed using three main terms and their variations. The search string is:

("recommender system" OR "recommendation system") AND ("LLM" OR "large language model") AND ("agent" or "agentic")

As expected, including the term "agentic recommender system" is not effective because of the novelty of this research field. Thus, we use the term "recommender system" with additional qualifiers. The term "recommendation system" is a common variant. We employ the term "LLM", and its variant "large language model" to refer to the recent trend in the field. However, these two terms, and their variations, alone are insufficient to capture the intended scope. We then specify the "agent" or "agentic" term, to specify the type of recommender system under investigation. Given the novelty of the term agentic AI, we instead use the broader terms "LLM" and "agent" to ensure sufficient coverage. We do not target recommender systems grounded in the traditional literature on multi-agent systems [10] and AI agents [9].

The search string remains relatively simple, as strict formulations are impractical given the field's emergence. For example, we seek recommender systems with multiple agents. Additional terms such as "multi-agent" or "multi-LLM agent" may be included. But this reduces the number of studies returned as there is no definite consensus regarding the terms in the field.

The steps to inspect each study returned are straightforward:

1) We first read the title and look for indications of LLM agents. In some cases, the title alone is sufficient.
2) If the title does not make it clear, then we read the abstract and seek any indication that the proposed recommender system includes LLM agents.
3) If it is still not clear, we move to the main section of the study. We inspect figures and key paragraphs that describe the proposed recommender system to identify indications of LLM agents.
4) If none of these efforts are fruitful, we then examine the entire paper to determine the presence of LLM agents.

In the end, if it remains unclear whether LLM agents are included in the recommender system proposed in the study, we exclude the paper from the review and document the reason for exclusion.

Once the paper is approved, it is subjected to inclusion and exclusion criteria, detailed in the next subsection. For the papers included in the review, we read them once again seeking information relevant to each research question. This information is detailed in the following subsection. We record this information and prepare it for analysis.

5) Inclusion and Exclusion Criteria: We outline the criteria used to exclude returned studies. The number of studies for each criterion is shown in a subsequent subsection. The exclusion criteria are presented as follows:

1. Accessibility
   - 1.1. No access
   - 1.2. Not in English
2. Publication
   - 2.1. Workshop proposal
   - 2.2. Conference proceedings
   - 2.3. Tutorial description
   - 2.4. Research proposal
   - 2.5. Perspective paper
   - 2.6. Position paper
   - 2.7. Thesis
   - 2.8. Book
   - 2.9. Survey of the literature
   - 2.10. Review of the literature
3. Alignment
   - 3.1. Not in the recommendation domain
   - 3.2. No recommendation framework
   - 3.3. Not agentic
   - 3.4. Dataset description
   - 3.5. Reinforcement learning agent
   - 3.6. User simulation
   - 3.7. Simulation framework
   - 3.8. Recommender system security
   - 3.9. Recommendation explanation
   - 3.10. Recommender system assessment

We exclude studies that we do not have access to, despite efforts, and those not written in English, which is a common practice in systematic literature reviews. We exclude workshop proposals, conference proceedings, or tutorial descriptions, as well as research proposals, perspective papers, and position papers, as they lack sufficient substantive information for inclusion. Theses and books are excluded because they often address either overly broad or highly specific dimensions of a topic and the substantial time for analysis. Other surveys or reviews of the literature are excluded from analysis but presented for further research. Lastly, studies that are not in the recommendation domain, or that do not describe a framework, or do not propose an agentic RS, or focus on other aspects of recommender systems, are also excluded. We, however, do not limit the year of publication.

[3]https://www.scopus.com/
[4]https://www.webofscience.com/
[5]https://dl.acm.org
[6]https://ieeexplore.ieee.org/
[7]https://arxiv.org
[8]https://scholar.google.ca

TABLE II
INFORMATION EXTRACTED FROM EACH STUDY TO ADDRESS THE RESEARCH QUESTIONS

RQs	Data Items
Info	Publication title
	Framework name
	Venue of publication
	Publication type (e.g. journal article)
	Year of publication
	Search engine that index the study
	Repository
RQ1	Agent names
RQ2	Agent roles
RQ3	The LLM model
RQ4	The knowledge augmentation strategy (e.g. RAG [45])
RQ5	Connections between agents
RQ6	Persona
	Instruction
	Input data
	Context (e.g. other agents)
	Steps
	Cues (e.g. requirements or limitations)
	Output format
	Examples
RQ7	Prompt strategy (e.g. CoT [46])
RQ8	Reported integrations (e.g. database)
RQ9	Reported use cases (e.g. direct recommendation)
RQ10	Recommendation strategy (e.g. collaborative filtering)
RQ11	Datasets and domains
RQ12	Metrics
RQ13	Baselines
RQ14	Metric values
RQ15	Reported research gaps

6) Extracted Data Items: Here, we outline the information extracted from each study that are used to address each research question. This information is presented in Table II.

B. Demographics

We present the results of the search string, the exclusion criteria, and the final list of studies selected for review. We also discuss the demographic characteristics of the included studies and the trends observed in this research field.

Our search string, as explained in the previous section, was adapted for each of the four search engines included in this review. Table III reports the number of studies retrieved from each search engine. The query was executed at the beginning of September 2025.

We removed duplicates and obtained a total of 140 studies. These studies were subjected to the exclusion criteria. Table IV details the reasons for exclusion of each study based on the exclusion criteria discussed previously.

Of the initial 140 initial studies, 44 were selected for review. Several studies are not in the recommendation domain, do not describe a framework for a recommender system, or employ LLM agents solely for user simulation. However, they mention recommender systems in their abstracts as a potential assistant and are retrieved.

Two notable rows in Table IV correspond to the surveys and reviews excluded by the criteria. Although they are not reviewed here, they may provide useful insights for readers, since they are related to the research field of ARS. For this reason, they are listed in Table V. A deeper analysis of these studies and the contextualization of this systematic review among these surveys and reviews is presented in Section II-D.

TABLE III
NUMBER OF STUDIES RETRIEVED FROM EACH SCIENTIFIC SEARCH ENGINE

Search Engine	Number
Scopus[9]	75
Web of Science[10]	13
ACM Digital Library[11]	11
IEEE Xplore[12]	4
arXiv[13] & others (indexed by Scopus)	84
Total (after removing duplicates)	140

TABLE IV
EXCLUSION CRITERIA APPLIED AND TOTAL NUMBER OF STUDIES INCLUDED IN THE SYSTEMATIC LITERATURE REVIEW

Criteria	Number
Total (after removing duplicates)	140
No access	0
Not in English	2
Workshop proposal	2
Conference proceedings	1
Tutorial description	1
Research proposal	1
Perspective paper	1
Position paper	2
Thesis	1
Book	1
Survey of the literature	6
Review of the literature	3
Not in the recommendation domain	11
No recommendation framework	9
Not agentic	7
Dataset description	2
Reinforcement learning agent	4
User simulation	9
Simulation framework	7
Recommender system security	7
Recommendation explanation	1
Recommender system assessment	2
Total (after exclusions)	44

We present a demographic analysis of the 44 studies included in this review. We start with the nature of the agentic recommender systems. Table VI shows the results. The results indicate the dominance of multi-agent systems over others, showing a trend toward collaboration between agents in recommender systems, instead of a single agent handling all requests. The main reason is the need for specialized agents in different tasks driven by the volume of data and the goal of personalization in recommender systems. In addition, agent tools (e.g. algorithms, libraries) are more relevant to single-agent recommender systems, given the lack of reasoning agents to perform tasks.

Next, we analyze the 44 studies based on their type: journal article, conference paper, or arXiv. Although the types are straightforward, one clarification is necessary. In some cases,

[9]https://www.scopus.com/
[10]https://www.webofscience.com/
[11]https://dl.acm.org
[12]https://ieeexplore.ieee.org/
[13]https://arxiv.org

TABLE V
SURVEYS AND LITERATURE REVIEWS RETRIEVED IN THE SEARCH BUT EXCLUDED BY THE SYSTEMATIC REVIEW CRITERIA

Title	Reference	Type
Large Language Models for Information Retrieval: A Survey	[32]	Survey
Survey for Landing Generative AI in Social and E-commerce Recsys – the Industry Perspectives	[47]	Survey
A Survey of Large Language Model Empowered Agents for Recommendation and Search: Towards Next-Generation Information Retrieval	[39]	Survey
A Survey of Personalized Large Language Models: Progress and Future Directions	[33]	Survey
A Survey on LLM-powered Agents for Recommender Systems	[40]	Survey
Safety at Scale: A Comprehensive Survey of Large Model Safety	[31]	Survey
All Roads Lead to Rome: Unveiling the Trajectory of Recommender Systems Across the LLM Era	[48]	Review
A Review of Large Language Models: Fundamental Architectures, Key Technological Evolutions, Interdisciplinary Technologies Integration, Optimization and Compression Techniques, Applications, and Challenges	[29]	Review
AI generations: from AI 1.0 to AI 4.0	[49]	Review
AgentSociety Challenge: Designing LLM Agents for User Modeling and Recommendation on Web Platforms	[50]	Review

TABLE VI
NATURE OF AGENTIC RECOMMENDER SYSTEMS PROPOSED ACROSS THE REVIEWED STUDIES

Nature	Number
Multi-agent + tools	4
Multi agent	22
Single agent + tools	12
Single agent	6

TABLE VII
TYPE OF THE STUDIES

Type	Number
Journal Articles	5
Conference papers	21
arXiv (only)	18

TABLE VIII
DISTRIBUTION OF STUDIES BY YEAR OF PUBLICATION

Year	Number
2025	26
2024	15
2023	3

it is common practice to deposit manuscripts on arXiv prior to formal publication. To avoid confusion, in this table, journal articles are classified as such even if they also appear on arXiv, and conference papers are classified as conference papers even if they also appear on arXiv. As a result, the nine preprints listed in the arXiv row are not returned as peer-reviewed publications by the search engines. The types and the results can be seen in Table VII.

We next analyze the 44 studies by type: journal article, conference paper, or arXiv. Results can be seen in Table VII. The longer review cycle accounts for the lower number of journal studies. Furthermore, the greater number of conference papers enhances the quality of the analysis of this systematic review, as these papers are peer-reviewed.

We also analyze the distribution of studies by year of publication. Table VIII reports the number of studies per publication year.

The emergence of the field of agentic recommender system is evident from the absence of publications prior to 2023. The numbers for 2025 are limited to the period January - September, when the search query was executed. Nevertheless, the number of publications nearly doubled from 2024 to 2025, indicating a clear trend toward increased output and reflecting growing interest in ARS that is expected to continue in 2026 and beyond.

We have previously classified studies by type (e.g. journal articles and conference papers), but we did not present the specific journals or conferences. We now present these publication venues, as they may be of interest to some readers. Table IX reports the venues along with the number of studies identified in each.

The three studies from the ACM Web Conference address ARSs in the context of a conference challenge on recommender systems. ACM SIGIR contributes three studies as well, but these are listed separately due to the year of the conference. The preprint repository arXiv is placed at the bottom to prioritize peer-reviewed venues. Several venues are associated with recommender systems, software engineering, artificial intelligence, big data, information retrieval, the internet of things (IoT), linguistics, and social systems, indicating the main areas connected to agentic recommender systems.

Finally, we present the set of 44 studies included in this review. These studies are presented in Table X.

The table lists each study in a separate row. Each row begins with the name of the agentic recommender system framework proposed in the study. For studies that do not provide a name, one is assigned based on the first author or by analogy with the proposal. The table then presents the study title, truncated for space limitations, along with the year of publication. The nature and the type of each study are summarized single-letter labels: under "Nature", M denotes multiple LLM agents and S denotes single agents, regardless of the presence of tools; under "Type", J denotes journal articles, C denotes conference papers, and P denotes arXiv preprints. The last column discloses the repository, when reported in the study, allowing readers to access the code or related resources. The order of the studies in the table has no particular meaning.

This concludes the demographic analysis. The 44 studies have been read following the protocol discussed in the previous subsection, with the results presented in the next subsection.

C. Results

This subsection presents the results of the systematic literature review on agentic recommender systems.

TABLE IX
PUBLICATION VENUES OF THE STUDIES

Venues	Number
34th ACM Web Conference, WWW Companion 2025	3
48th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2025	2
IEEE Transactions on Computational Social Systems	2
47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024	1
37th Conference on Neural Information Processing Systems, NeurIPS 2023	1
18th ACM Conference on Recommender Systems, RecSys 2024	1
Proceedings of the Nineteenth ACM Conference on Recommender Systems	1
2024 IEEE International Conference on Big Data, BigData 2024	1
2023 Findings of the Association for Computational Linguistics: EMNLP 2023	1
2024 Findings of the Association for Computational Linguistics: NAACL 2024	1
29th Pacific-Asia Conference on Knowledge Discovery and Data Mining, PAKDD 2025	1
30th Americas Conference on Information Systems, AMCIS 2024	1
30th International Conference on Intelligent User Interfaces Companion, IUI 2025	1
33rd ACM Web Conference, WWW 2024	1
33rd Companion of the ACM World Wide Web Conference, WWW 2023	1
47th European Conference on Information Retrieval, ECIR 2025	1
4th International Conference on Computer Technologies, ICCTech 2025	1
5th IEEE Annual World AI IoT Congress, AIIoT 2024	1
7th Conference on Conversational User Interfaces, CUI 2025	1
ACM Transactions on Information Systems	1
Computers in Biology and Medicine	1
Scientific Reports	1
arXiv	18

1) RQ1: We have identified 112 LLM agents and 48 tools in 44 studies. This averages at 2.5 agents and about 1 tool per proposal. Table XI lists LLM agents of five agentic RSs. The icon beside their names represents their role, which is detailed in RQ2.

a) Agentic recommender systems are multi-agent: A clear trend is the adoption of a multi-agent approach. In general, proposals do not rely on a single LLM to perform all tasks in the recommendation process, favoring a range of LLM agents specialized in specific tasks, facilitated by customization methods such as fine-tune, RAG, or ICL. The use of multiple LLM agents improves modularization and personalization, which are very important characteristics of RSs. Some studies have LLM agents for each dataset [75] or participating in voting [77] or in a discussion [87]. Among multi-agent proposals, the average number of agents rises to 3.6, while the average number of tools drops to just 0.3, further supporting a multi-agent, no-tools strategy and greater reliance on agent capabilities. Despite some proposals having 5 or even 11 LLM agents described, an average of 3 or 4 agents is explained by the lower performance that actual agents introduce in the process. A not high and not low number of agents on average is the balance between accuracy and performance.

b) How to name an LLM agent? Most studies assign names to their agents according to their functional roles, as this facilitates ease of recall. For instance, agents may be designated as “Item Agent”, “Reflector”, or “Searcher”. In contrast, single-agent systems are typically left unnamed, since the agent is generally presumed to be synonymous with the framework itself.

c) Agents are collaborative and internal. Anthropomorphic too? In general, LLM agents collaborate in the recommendation process. The exception is the “Reflector” agents, which usually check the recommendation list for correctness before presentation. This result reflects a human trend of developing components that work in harmony, despite competition having its benefits. One example is generative adversarial networks [94]. A study of the competition of agents in recommender systems remains an open research area. Moreover, only one proposal investigated anthropomorphic characteristics of the LLM agent [93], such as giving it a name or communicating with the user in an emotionally closer way. Other authors rely on behavior described in the prompt to tailor the LLM agent’s natural language response. As LLMs in general and even RSs become more personal, this remains an open question and research avenue.

RQ1

Q: What agents are used in agentic recommender systems?
A: Table XI shows the agents used in four studies.

- Agentic recommender systems are typically multi-agent, with an average of 3.6 agents, and employ relatively few tools while seeking to balance accuracy, personalization, and performance.
- The potential benefits of agent competition in recommendation generation is not explored.
- The use of anthropomorphic characteristics in LLM agents used in ARSs may engage the user in novel ways and remains an unresolved research question.

2) RQ2: We have identified 13 unique agent roles, which are shown in Table XII. A discussion of the results follows.

a) LLM agents recommend or assist a recommender: LLM agents take part in the recommendation process either as the actual recommender or in an auxiliary role, such as coordinating agents or handling user or item data. Some authors

TABLE X
LIST OF STUDIES INCLUDED IN THE SYSTEMATIC LITERATURE REVIEW

Framework Name	Ref	Title	Year	Nature	Type	Repository
DCICDRec	[51]	Harnessing Generative Large Language Models for …	2025	S	A	Not reported
RAH!	[52]	RAH! RecSys-Assistant-Human: A Human- …	2024	M	A	https://github.com/TheRepoForRAH/RAH
Hybrid-MACRS	[53]	A Hybrid Multi-Agent Conversational Recommen …	2024	M	C	Not reported
AgentCF	[54]	AgentCF: Collaborative Learning with Autonomou …	2024	M	C	https://github.com/RUCAIBox/AgentCF
AgenticFramework	[24]	An Agentic AI-based Multi-Agent Framework for …	2024	M	C	Not reported
SmartApp-Vision	[55]	Large Language Model-Driven Immersive Agent	2024	S	C	Not reported
MACRec	[56]	MACRec: A Multi-Agent Collaboration Framewor …	2024	M	C	https://github.com/wzf2000/MACRec
RecAI	[57]	RecAI: Leveraging Large Language Models for …	2024	S	C	https://github.com/microsoft/RecAI
RecMind	[58]	RecMind: Large Language Model Powered Agent …	2024	S	C	Not reported
CORE	[59]	Lending Interaction Wings to Recommender Syste …	2023	S	C	Not reported
SalesOps	[60]	Salespeople vs SalesBot: Exploring the Role of …	2023	S	C	https://github.com/salesforce/salesbot
MACRS	[61]	A Multi-Agent Conversational Recommender System	2024	M	P	Not reported
AFL	[62]	Agentic Feedback Loop Modeling Improves Reco …	2025	M	C	https://github.com/Lanyu0303/AFL
ChatCRS	[63]	Incorporating External Knowledge and Goal Guid …	2024	M	P	https://github.com/lichuangnus/ChatCRS
LIKR	[64]	LLM is Knowledge Graph Reasoner: LLM's …	2025	S	C	Not reported
MAS4POI	[65]	MAS4POI: a Multi-Agents Collaboration System …	2025	M	C	https://github.com/yuqian2003/MAS4POI
Thakkar	[66]	Personalized Recommendation Systems using Mul …	2024	M	P	Not reported
Rec4Agentverse	[67]	Prospect Personalized Recommendation on Large …	2024	M	P	https://github.com/jizhi-zhang/Rec4Agentverse_Case
AsyncMLD	[68]	AsyncMLD: Asynchronous Multi-LLM Framewor …	2023	S	P	Not reported
InteRecAgent	[69]	Recommender AI Agent: Integrating Large Langu …	2025	M	A	https://aka.ms/recagent
AgentCF++	[70]	AgentCF++: Memory-enhanced LLM-based Agen …	2025	M	C	https://github.com/jhliu0807/AgentCF-plus
TAIRA	[71]	Thought-Augmented Planning for LLM-Powered …	2025	M	P	https://github.com/Alcein/TAIRA
MATCHA	[72]	MATCHA: Can Multi-Agent Collaboration Build a …	2025	M	P	Not reported
MMAgentRec	[73]	MMAgentRec, a personalized multi-modal recom …	2025	M	A	Not reported
SmartEats	[74]	SmartEats: Investigating the Effects of Customizab …	2025	S	C	Not reported
RAMaR	[75]	Retrieval Augmented Multi-agent Recommender	2025	M	C	Not reported
DummyAgent	[76]	Intelligent Agents with Adaptive Knowledge Fusio …	2025	S	C	https://github.com/wzf2000/DummyAgent
RecHackers	[77]	Personalized Recommendation Agents with Self- …	2025	S	C	Not reported
MedicationAssistant	[78]	An adaptive language model-based intelligent med …	2025	S	A	Not reported
ProactiveAgent	[79]	Managing Proactivity in E-Commerce Interfaces: …	2025	M	C	Not reported
FreudianRS	[80]	Integrating Freudian Psychological Constructs - Id, …	2025	S	C	Not reported
RecSys-Agent	[81]	Large Language Model-based Recommendation S …	2025	S	C	https://github.com/tommasocarraro-sony/recsys-agent
ARAG	[82]	ARAG: Agentic Retrieval Augmented Generation …	2025	M	P	Not reported
CSI	[83]	Towards Personalized Conversational Sales Agents: …	2025	S	P	Not reported
i2Agent	[84]	iAgent: LLM Agent as a Shield between User and …	2025	M	P	https://github.com/agiresearch/iAgent
Zoom	[85]	Multi-agents based User Values Mining for Recom …	2025	M	P	Not reported
RuleAgent	[86]	RuleAgent: Discovering Rules for Recommendatio …	2025	S	P	Not reported
DecisionGPT	[87]	Agentic LLM Framework For Adaptive Decision …	2025	M	P	https://github.com/HydroComplexity/DecisionGPT
HapticAgent	[88]	Leveraging LLMs to Create a Haptic Devices' Rec …	2025	M	P	Not reported
AgenticRAG	[89]	Developing an Artificial Intelligence Tool for Pers …	2025	M	P	Not reported
HiRMed	[90]	Tree-based RAG-Agent Recommendation System: …	2025	M	P	Not reported
User-as-a-tool	[91]	Insert-expansions for Large Language Model Agents	2024	S	C	Not reported
GenUP	[92]	GenUP: Generative User Profilers as In-Context …	2024	M	P	https://github.com/w11wo/GenUP
Sunnie	[93]	"I Like Sunnie More Than I Expected!": Exploring …	2024	S	P	Not reported

TABLE XI
AGENTS AND THEIR ROLES REPORTED IN FIVE STUDIES

MAS4POI [65]	MACRec [56]	RAH! [52]	MACRS [61]	ChatCRS [63]
Manager	Manager	Act	Planner	Conversational
User agent	Task Interpreter	Learn	Responder (Ask)	Knowledge Retrieval
Data agent	User Analyst	Perceive	Responder (Recommend)	Goal Planning
Searcher	Item Analyst	Critic	Responder (Chat)	
Reflector	Searcher	Reflect	Feedback	
Analyst	Reflector	Personality Library		
Navigator				

TABLE XII
UNIQUE AGENT ROLES DESCRIBED IN THE STUDIES OF THIS SYSTEMATIC REVIEW

Agent	Role	Description
Manager		Has a central role and coordinates execution.
Recommender		Generates a recommendation.
Interpreter		Structures user input for the Manager agent.
Gatekeeper		Prevents malicious prompt attacks and considers the ethical aspects of prompts.
Goal Planner		Identifies the user goal. This will be used as the recommendation task.
User Agent		Handles user information. It may identify the user or search for others with similar profiles.
Item Agent		Handles item information. It may provide item information or search for similar ones.
Domain-Specific Agent		Provides domain-specific functionalities, based on the domain of the recommender system.
Searcher		Search other datasets (e.g. internet) for information.
Tools		Algorithms or other systems. Tools are not agents and are included for completeness.
Reflector		Checks the correctness of the output message.
Responder		Tailors the output prompt to the user by using appropriate language based on the user.
Feedback Agent		Handles user feedback. This may relate to the recommendation or the recommender system.

explore the recommendation abilities of LLMs by asking it to rank a recommendation list presented in the prompt based on user characteristics, while other authors often augment the recommender agent knowledge with strategies (e.g. knowledge graph) or additional information (e.g. a search of item descriptions). In most cases, the recommendation is usually performed by a single agent, opening up opportunities for studies that explore horizontal or vertical hierarchies of agents for recommendations, such as the tree-structure used in HiRMed [90], which progressively improves recommendations based on department and then specialty in the medical field.

As expected, Table XII includes a Manager agent, which coordinates agent activities, and a Recommender agent that calculates or generates recommendations. In several studies, both roles are played by the same LLM agent, whereas other studies, such as CORE [59], use a traditional RS.

Other notable agents are the User and Item Agents, which handle user and item data, and the Reflector, which critiques the recommendation before it is presented to the user. MATCHA [72] is the only study proposing agents for data and the ethical safety of the system. In fact, seven studies are excluded from this review (Table IV) because they investigate malicious attacks on RSs, not recommendations, showing that agentic RS security is an emerging major research area.

b) Tools represent agent roles: One of the agents' roles in Table XII is Tools. This entry is reserved for entities that are important for the recommendation process in a study, but do not qualify as an LLM agent. For example, LIKR [64] uses a knowledge graph for user and item storage and handling. The most common tools are (i) ranking algorithms, (ii) a knowledge graph or database, and (iii) some specialized ones, such as mathematical tools or mapping utilities. For performance reasons and depending on the type of ranking (e.g. re-ranking), the ordering of candidate items can be done faster with the aid of a tool. Some knowledge graphs or vector databases contain or encode strategies that can be used by the agent for more accurate results. Lastly, given the difficulties LLMs have in handling advanced mathematics [95], [96] and the additional time to specialize agents to perform some domain-specific tasks, authors sometimes resort to tools for calculations or some specialized abilities, such as reading maps.

c) The agent role list provides a starting point: One last insight in RQ2 is that the list of unique agent roles shown in Table XII can be used in early discussions in agentic recommender system development. Although this list of agent roles is extensive, not all roles are required in an agentic RS and agents may perform more than one role. The list still assists developers in choosing the right roles for the agentic recommender system being designed.

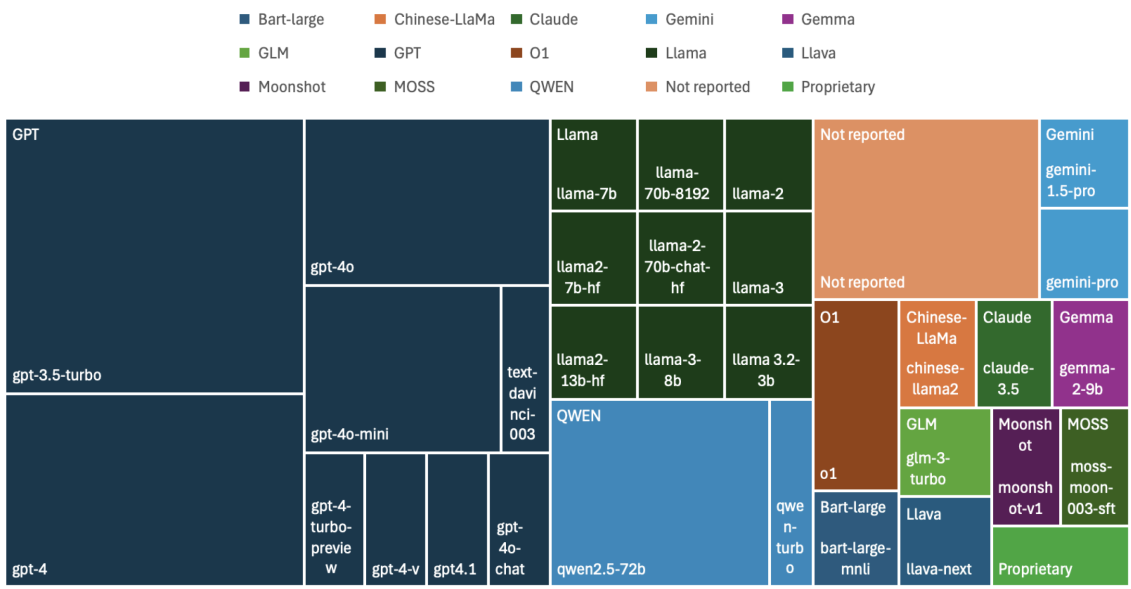

Fig. 1. LLMs used in agentic recommender systems included in this study.

RQ2

Q: What roles do agents used in agentic recommender systems play?
A: Table XII shows the 13 roles.

- Most studies include either a Manager or a Recommender agent, often both in the same LLM agent.
- Hierarchical recommendations remains unexplored.
- Agentic RS security is a growing area of research, both in the system and in the ethical aspects.
- Tools remain for very specialized tasks, such as strategies or calculations, as adapting LLM agents can be time-consuming.
- The extensive list of agent roles shown in Table XII is a starting points for development.

3) RQ3: We have identified 59 uses of LLMs as agents in the proposed agentic RSs, including one proprietary model. Five studies did not report the LLM used. We analyze LLMs used in all agent roles within the recommendation process, not only the recommender. LLMs reported in ablation studies are not considered. Fig. 1 illustrates our discussion.

a) GPT is popular: More than half of studies (31) use a version of the GPT model as an LLM agent. The main reasons for this result are a simplified API, popularity, and good accuracy. However, there is no clear winner among GPT models, such as GPT 3.5, 4, 4.1, or 4o. Llama models are a runner-up. They are used in 9 studies with the assistance of Hugging Face's[14] libraries, which simplifies execution. While GPT models are executed remotely, Llama models are local, raising research questions about the tradeoff between performance and security.

b) Less used models and niche models are presented: Few studies included Google's Gemini or Anthropic's Claude models. Other popular models are not explored, such as DeepSeek, Mistral, or Perplexity. The studies describe no apparent reason, ruling out technical difficulties. The result may be explained by low popularity or price. The performance of these models in agentic recommender systems remains an open research area.

c) General models over specially trained ones: Few studies use "-chat" or "-instruct" models as their LLM agents. The reasons are twofold. First, most agents are internal, leaving "-chat" models only to interface with the user. Authors prefer tool-using or function-calling LLMs for their reasoning capabilities, while "-instruct" models are left for text summarization tasks such as goal extraction.

d) Agents are pros: There is a clear preference for advanced LLMs in agentic recommender systems, evidenced by the "turbo" or "pro" versions and number of parameters, often over 70 billion. They are used especially in coordination or recommendation agents for their accuracy. However, a study of the best LLM for each task is still lacking.

e) Recommenders should talk and see: HapticAgent [88] uses an image model for recommendation, but text recommendations still dominate the research field. Multi-modal recommendations, taking into consideration data beyond text, such as images, audio, or video, are still unexplored and remain an open area of research.

[14] https://huggingface.co

TABLE XIII
THE KNOWLEDGE AUGMENTATION STRATEGY USED IN THE FRAMEWORKS

Strategy	Number
ICL	16
RAG	9
Fine-tune	5
None	5
Not reported	11

RQ3

Q: What large language models (LLM) are used in agentic recommender systems?
A: Fig. 1 identifies 59 uses of LLM models.

- GPT leads the way, followed by Llama. The former executes remotely and the latter, locally. The trade-off between performance and security remains to be studied.
- Little is known about the performance of less popular or niche models. More studies required.
- Tool-using and function-calling models are preferred over chat or instruct models.
- Advanced models are preferred for recommendation tasks, such as turbo, pro, or over 70 billion parameters. More studies on the applicability of less powerful models are required.
- Multi-modal recommendations remain unexplored. Recommendations based on image, audio, and video may enhance user experience.

4) RQ4: We summarize the main knowledge augmentation techniques used in the studies when reported. The details are provided in the following paragraphs.

a) Agents learn in-context: We have found three main techniques for agent knowledge augmentation: in-context learning (ICL) [97], retrieval augmented generation (RAG) [45], and fine-tuning [98]. Table XIII shows the frequency at which each technique is reported and highlights the primacy of ICL. A brief definition of each technique is given below:

- ICL [97]: In-context learning refers to providing data to an LLM using the prompt.
- Fine-tuning [98]: When fine-tuning an LLM, new documents or data follow the training pipeline and are added to the vocabulary and knowledge of the LLM.
- RAG [45]: In retrieval augmented generation, new documents or data are processed through the training pipeline and added to the vocabulary and knowledge of the LLM at inference time. A more flexible RAG also uses relational databases.

The main reason for the result in Table XIII is the simplicity of prompt usage and advancements in prompt context windows. Candidate lists, user profiles, conversation history, and sometimes even the history of interactions are passed to the agent through the prompt. ARAG [82] summarizes historical interactions, which is more efficient. More details about prompt input data are discussed in RQ6.

No study reported the adoption of Model Context Protocol (MCP), mainly because it would be used in the agentic framework supporting the creation and execution of agents. JSON files are commonly used in data passing, either between agents or as the response from another system. However, both MCP answers and JSON files are usually added to the prompt and identified as ICL. Some studies use SQL to query relational datasets. However, queries are usually hardcoded in a function, not generated during execution, and the database answer is copied to the prompt by the main program, which therefore counts as ICL. Studies on the use of MCP, automatic JSON processing and SQL query generation during execution may enable new knowledge augmentation strategies.

In Table XIII the label "None" refers to studies that opted to use the LLM's own knowledge to provide recommendations. This can be done for movies or other well-known items, but it is limited.

b) There is a need for improved ways to report strategies: Unfortunately, several studies either do not report the knowledge augmentation strategies used for the agents or do not do so clearly. As the field of agentic AI evolves, more robust ways of reporting scientific research are required, that ensure relevant and complete disclosure and explanation of how AI agents or LLMs are used.

Moreover, OpenAI's API assists developers in fine-tuning models, using RAG, or passing data in ICL. This challenges efforts to capture the strategies when authors do not report them explicitly. This also reveals a lack of frameworks for agent knowledge augmentation strategies that can assist developers in this process.

RQ4

Q: What knowledge augmentation strategies are used in agents used in agentic recommender systems?
A: Table XIII presents three main strategies.

- ICL is the top choice.
- MCP, JSON, and other alternatives are not explored because of the reliance on an underlying framework, opening up research avenues.
- Some authors use the LLM's own knowledge about items, such as movies, which merits further study.
- Authors should report knowledge augmentation strategies to ensure better analysis results.
- There is a lack of knowledge augmentation frameworks for LLM agents, resulting in most authors relying on OpenAI's APIs.

5) RQ5: The relationships among LLM agents in the 44 studies are analyzed based on the reports. The paragraphs below explain the conclusions.

a) Manager in the center: Most studies follow a Manager-centric layout of agents. The Manager coordinates all other agents and message passing. In many studies, the Manager also plays the role of the Recommender, as in ChatCRS [63], except when a traditional recommender is used, as in CORE [59]. Two other authors used a different layout strategy. RecHackers [77] uses a prompt-voting mechanism, where different prompts are generated and voted on to generate the best recommendation results. The same idea can be applied to the results of different agents, in a strategy similar to the ensemble recommendations. DecisionGPT [87] generates different agents with different roles and encourages discussion among agents to generate the best recommendations. Novel strategies may consider hierarchies or a mix of the layouts discussed.

b) Where are the orchestration frameworks? The use of agent orchestration frameworks is still limited. Very few studies have used frameworks such as LangChain[15] or CrewAI[16]. Authors often rely on the programming language to call agents. The same situation is observed in message passing, where agent responses are copied into the next agent's prompt with the assistance of the programming language. More sophisticated approaches based agentic frameworks to support agent orchestration and message passing need to be investigated. Simplified APIs and graphical user interfaces can assist developers in adopting these frameworks.

c) Some roles remain unexplored: Among most studies analyzed, the efforts in improving the *accuracy* of recommender systems result in an oversight of other agent roles, such as the Searcher, Reflector, or the Gatekeeper, because of the tradeoff with other non-functional requirements, such as performance or security. Proposals that include a wider range of agents and address other requirements are promising.

d) A generalized framework: Fig. 2 shows a generalized framework for agentic recommender systems. The framework is derived from all studies analyzed and includes a broad set of cases and most relationships. We briefly describe a workflow and discuss its variations.

1. Initially the user provides a prompt with a recommendation request, which is read by the Interpreter.
2. The Interpreter formats the message appropriately and contacts the Manager.
3. The Manager first contacts the Gatekeeper to check for potential threats in the prompt and then contacts the Goal Planner to identify the user goal (e.g. direct recommendation, feedback, or recommendation explanation).
4. The Manager may contact other agents to gather information prior to generating a recommendation, including the User Agent, Item Agent, Searcher, Domain-Specific Agent, or external tools.
5. The Manager compiles the information gathered and passes it to the Recommender for generation, if a distinct agent is used.
6. The recommended list is passed by the Manager to the Reflector for a correctness check.
7. The final recommended list is sent to the Responder for a natural language presentation.

[15]https://www.langchain.com
[16]https://www.crewai.com

Some variations apply. First, user feedback can be handled by a specialized agent, such as RAH!'s [52] Reflect, or passed to Item and User agents for processing, such as in MACRS [61]. The real difference is in the number of agent relationships and the abilities of each agent to update its knowledge base. Secondly, the Gatekeeper may read the message before any agent to prevent vulnerabilities. Thirdly, Tools can be accessed by any agent to perform their tasks, resulting in many more relationships. Fourthly, in the studies, a single agent performs more than one of these roles (e.g. Manager and Goal Planner), thus reducing the number of relationships. Lastly, human-in-the-loop, multi-modal recommendations, cyber-physical recommenders, psychological personalization approaches, such as in FreudianRS [80], and other variations increase the complexity of the framework but may uncover novel and interesting relationships.

RQ5

Q: How to characterize agent relationships in agentic recommender systems?
A: Fig. 2 presents a framework constructed from agent relationships found in the studies.

- Most studies follow a Manager/Recommender-centric layout, despite the existence of voting or discussion layouts.
- Few authors use orchestration frameworks and message passing occurs by adding a variable to the next prompt.
- Other assistive agent roles, such as the Searcher, the Reflector, and the Gatekeeper, may assist in non-functional requirements beyond accuracy, such as performance or security.
- A generalized framework is presented and several variations are discussed.

6) RQ6: A key difference of this systematic literature review is a study of the prompts used in the proposed frameworks. Twenty-six studies disclose their recommendation prompts partially or fully in the manuscript or in a repository. AgentCF [54] is divided into three cases (basic, sequential and retrieval augmented) and RecMind [58] in two (direct and sequential), based on the recommendation type. InteRecAgent [69] reports the same repository as RecAI [57], thus its prompts are not reviewed. This increases the total to 28 cases. We divide this RQ into eight parts, based on the prompt part being analyzed. A ninth part presents and discusses a generalized prompt (Fig. 3) derived from the prompts analyzed.

a) Persona: More than half of the proposals analyzed (17 in total) specify a persona for the agent in the prompt. Descriptions typically fall into one of two categories, though some may belong to both. The first category is based on the function of the agent, as in "you are a recommender system" or a "shopping assistant". In some cases, this includes the type of the item involved, such as "a music recommender system". In other cases, it focuses on a domain of expertise, for example "a

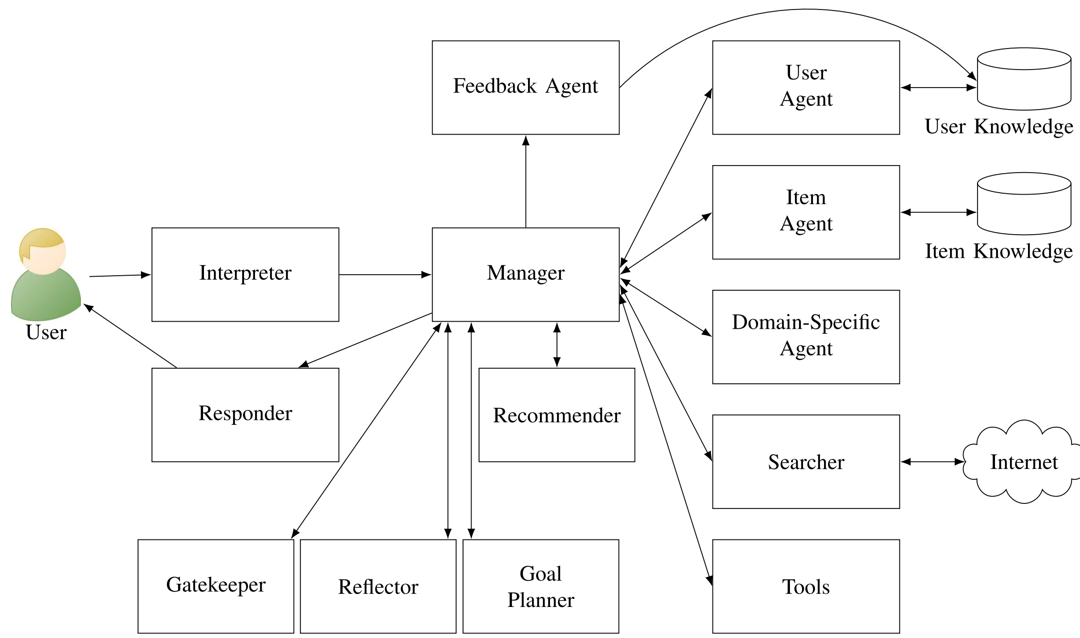

Fig. 2. A framework for agentic recommender systems. Agents are found in RQ2 and presented in Table XII.

medical expert". The second category is based on the behavior of the agent, such as "helpful", "compassionate", or "excellent". This second one is discussed in more detail in the paragraph about Cues, in item *f)* below.

The use of personas, especially domain experts, is correlated with better performance [99], but choosing the right persona or describing it correctly remains a challenge [100]. The impact of personas on the quality of recommendations is an open question. In addition, no study examines shifts in persona based on user characteristics or mood. For example, younger users may be more receptive to surprising recommendations.

b) Instruction: Almost all proposals include a description of the task in the form of an instruction to the LLM agent (26 in total). Common formats are "your task is..." or in imperative mode "generate recommendations for...", or both. One study, GenUP [92], uses a question format as in "Given [user history], what is the next POI?". Recsys-Agent [81] does not instruct the agent directly, but allows it to decide independently, and only provides some limitations on the choice.

Novel ways to instruct LLM agents are still under investigation and remain an open question. Few authors resort to questions, negative sentences, or rely on examples. A token representing a vector may be sufficient for recommendation generation. Moreover, the abilities of LLM agents to follow instructions still need improvement, as agents often hallucinate or act in unpredictable ways. Research on this topic is still early but promising [101], and may lead to advances in the field of agentic recommender systems [102].

c) Input Data: All but two proposals describe some input data in their prompts. Input data typically fall into one of the types shown in Table XIV.

Input data is often a string containing natural language text, for simplicity. However, some proposals resort to the JSON format for more structured messages or vectors, which is more efficient. The most common input data is "historical interactions" (12 in total), which is an essential part of recommender systems. Only two proposals use a scratchpad, indicating a lack of framework adoption for the orchestration of agents, since many frameworks require this variable for agentic loops.

A few proposals summarize inputs, such as the user profile or historical interactions, for efficiency. This can be done with the help of an LLM. Studies on the most distinctive features that need to be preserved in the summary, in the context of agentic RSs, are still unexplored. Some proposals include the item type or the dataset to access as parameters in the prompt, for customization. However, behavioral traits that can tailor the recommendation generation or output are not observed, opening up opportunities for more studies.

TABLE XIV
TYPES OF INPUT DATA

Input data	Description
Item type	E.g. movie, music, POI.
Conversation goal	E.g. direct recommendation.
Reflections	For correctness.
User identifier	Often the user id.
User description	E.g. age, preferences, etc.
Dataset	For access.
Target item	if rating prediction.
Target item information	Additional information.
Additional knowledge	For assistance, e.g. knowledge graph.
No. of candidate items	The number.
Candidate item list	The list.
Candidate items' info	Additional information.
User historical info	E.g. purchases, interactions.
Chat history	Historical messages.
Output format	Allows for personalization.
Scratchpad	For intermediate results.

Persona	You are the Recommender agent in an agentic recommender system ...
Instruction	Your task is to provide recommendations to the user based on the information available to you.
Input data	The user id is: ${user.id}. The user demographics are ${user.age}, ${user.sex}, ${user.demographics3}, ... The user preferences are [(${item.id},${user.rating})]. The user aversions are [(${item.id},${user.rating})]. The summarized chat with the user is ${user.chat}.
Context	You have access to the following list of agents: [(${agent.name}, ${agent.description})].
Steps	To provide a recommendation, follow these steps: 1. Identify users that are similar to the current user with respect to the items they prefer and those they dislike. 2. For those users, retrieve the five most preferred items and calculate the average rating. ... n. Provide the candidate list to ${agent.id} for a correctness check.
Cues	You should follow the steps outlined in order. You should not create new items. An item should be recommended only once in the candidate list.
Output format	The candidate list should be ranked in descending order of preference. The candidate list should have one item per line.
Examples	user.id: 27 user.age: 18 user.sex: male user.preferences: [(item1,10),(item2,9)] user.aversions: [(item10,2),(item11,1)] user.chat: "I am looking for three ..." agents: [("Reflector","Checks correctness of the candidate list."), ("Responder","Tailors the output to the user.")] candidate_list: 1. item100 2. item101 3. item102

Fig. 3. A generalized prompt derived from the prompts analyzed in this study.

d) Context: We call "context" the prompt item that describes the other agents or tools an LLM agent has access to. Only a few proposals (four in total) explicitly describe context in the prompt. Common information includes agent and tool names and descriptions, SQL tools, and available agent actions. A large number of prompts do not report context mostly because message passing is controlled either by the framework or by the "main" function, thus not requiring an explicit mention.

Agent description is still based on a short natural language paragraph. Open research areas explore improved ways of describing agents, including the use of output examples, for example. More advanced ways consider whether the next agent (or tool) is collaborative or competitive, a level of agent priority, performance considerations, or even cost of invoking it. This information assists LLM agents in their decisions.

e) Steps: Some proposals (eight in total) include a list of steps to guide the LLM agent during execution, but the majority rely on agentic loops or have a fixed workflow. Table XV summarizes the reported steps. The steps can be further enriched with information from other agents, such as recommendation strategies or domain-specific information.

Specifying steps is especially common and important in domains such as coding [103] and mathematical tasks [96], but it is not widely observed in agentic RSs, highlighting the need for more proposals and experimental studies on its efficacy. There is a need for more frameworks that guide the LLM agents through steps, treating them as workflows rather than mere tokens in a prompt. This remains an open question.

f) Cues: Cues are short instructions or requirements that either direct or limit the actions or outputs of LLM agents. Many proposals (10 in total) include multiple cues. We classified the cues in four categories: psychological, behavioral, recommendation-related and output format-related. Table XVI provides several examples for reference. Several cues aim to mitigate hallucinations, such as cues about making up answers,

TABLE XV
A GENERALIZED SET OF STEPS REPORTED IN THE PROMPTS

#	Steps	Description
1.	Greeting	To address the user when needed.
2.	Query user about recommendations	Assesses if users would like recommendations.
3.	Analyze user preference.	This includes information about the user and short descriptions.
4.	Analyze user historical interactions.	This includes purchases, likes or favorites. It may be in order, depending on the recommendation task.
5.	Analyze candidate list, rearrange it.	Depending on the recommendation task, a candidate list is provided for ranking.
6.	Recommend following output format	Instruct the agent to follow an output format.
7.	Closing	A message to encourage user or ask for feedback.

TABLE XVI
CUES INCLUDED IN THE PROMPTS OF ARSS

Types	Cues
Psychological	• Be gentle, polite. • Be concise, professional. • Be compassionate, supportive.
Behavioral	• Do not hallucinate. • Do not write code. • Do not make up answers.
Recommendation-related	• Provide reasons first and only then recommend items. • Use the candidate item list for recommendations. • Explanations should be short. • Only use relevant information from external knowledge.
Output format-related	• Limits on characters in response. • Size of output list • Do not output the analysis process

outputting the analysis process or "Do not hallucinate" instructions. This highlights the need for clearer guidance to ensure tasks are performed correctly.

The use of language cues for conversational agents has been addressed in the literature [104], though their efficacy and impact on the recommendation domain remains open for investigation. Experimental studies could focus on the effects of cues on personalization, agent decision, and hallucinations. Since cues guide the psychology and behavior of agents, a framework for dynamically adapting the personality of LLM agents is an interesting approach, supported by a repository of reusable personalities.

Cues are usually defined by the engineer in the prompt at the beginning of system execution and are fixed. An interesting approach allows the LLM agent to autonomously decide the (psychological) cues for a task. The impacts of this capability remain unexplored. Lastly, no study has included cues related to system security whether ethical or malicious. This is another open research avenue in agentic RSs.

g) Output Format: About three quarters (20 in total) of proposals include an output format in their prompts. Output formats are helpful for analysis as structured output is easier to compare or tailor. The most common items in the output formats analyzed are instructions to output one item per line, to include item ids, and to present explanations either before or after the list. The need for these instructions shows that recommendation lists are not the primary choice of output of LLM agents. Less common output items relate to the use of a JSON format and the order of additional information, such as advantages, user profile, or messages from other agents. Only two proposals use a fixed sentence for output. This assists developers with insights for their own output format specifications.

There are some open research areas related to output formats. One example is the definition of output format patterns, such as paragraphs, lists, or tables, to simplify prompt construction. There is a lack of metrics about the adherence of the generated output to the format specified in the prompt. Lastly, studies should investigate the differences between output formats for humans and for agents, as interpretability is judged differently by both parties.

h) Examples: Most proposals (25 in total) do not resort to examples to influence the reasoning and output of LLM agents. One interesting case is RecSys-Agent [81] that specifies many examples following what can be called a many-prompt strategy. The main reasons for this result are the reliance on the output format and the fact that the recommendation process is highly complex. Experimental studies on the use of examples in the domain of agentic RSs continue to be in demand.

The definition of examples is often ad-hoc. A study of the types of examples and their breadth provides a rationale for their use. Moreover, little is known about the use of the same examples in different tasks or domains, especially when coupled with an understanding of user preferences, mood, or personality. For example, music examples may give insights on movie recommendations.

i) A generalized prompt: Fig. 3 presents a prompt based on the most common items discussed in RQ6. This prompt represents a first draft for engineers developing agentic RSs. Variations include psychological characteristics defined through Persona or Cues, richer agent descriptions emphasizing priority or performance in Context, and conditional statements in Steps. In general, open research questions on prompt engineering for agentic RSs include the minimal prompt required for good recommendations, the design of prompt patterns based on these items, and the keywords that ensure a secure prompt.

RQ6

Q: How are prompts characterized with respect to their structure, such as persona, instruction, input data, context, steps, cues, output format, and the presence of examples?

A: Table XVII summarizes the main findings of each part of the prompt.

- Fig. 3 presents a generalized prompt as a reference.

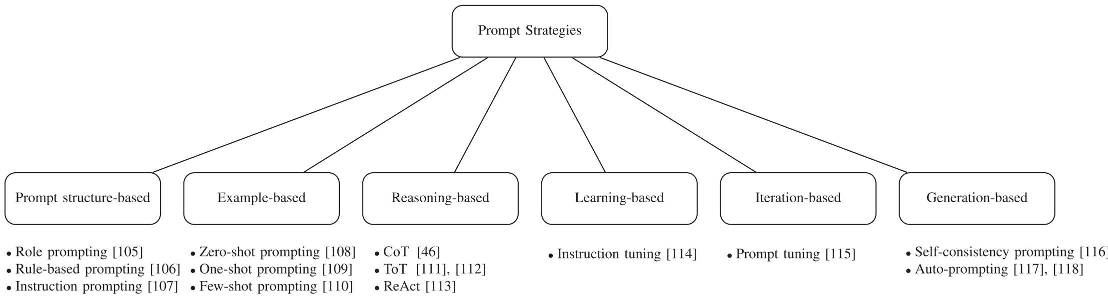

Fig. 4. An exploratory taxonomy resulting from the review of the literature on prompt engineering strategies.

TABLE XVII
A SUMMARY OF THE FINDINGS OF RQ6

Part	Findings
Persona	• Included in most studies. • Impacts on recommendation quality and changes based on user mood remain unexplored.
Instruction	• Almost all studies include instructions. • Alternative or standardized ways, such as questions, negative sentences, or using a token can be investigated.
Input Data	• Passing historical interactions through the prompt is common. • Behavior traits or datasets as input data can dynamically change an LLM agent and are worth studying.
Context	• Very few studies describe a context. • Advanced ways of describing a context include agent (or tool) characteristics, priorities, performance and cost, and can be investigated.
Steps	• Most studies rely on agentic loops or have a fixed message passing flow, and do not specify steps. • Frameworks that guide LLM agents through steps remain an open research area.
Cues	• We have identified four types of Cues: Psychological, Behavioral, Recommendation-related, and Output format-related. • Dynamic psychological trait changes in LLM agents enable advanced personalization and remain an open question.
Output	• Very common in studies. • Structured output formats and metrics of adherence remain unexplored.
Examples	• Almost never used. • Studies on breadth of examples and whether examples are dependent on task or domain are opportunities for research.

7) RQ7: We classify prompts in the proposals based on six prompt engineering strategies, which are presented in Fig. 4. Several opportunities related to prompt engineering research for agentic RSs remain open and warrant further discussion, as outlined below.

a) Role + instruction prompting is the most popular: Specifying a role for the LLM agent to play and providing it with a task in the form of an instruction is the most common prompting strategy used in the proposals (12 in total), primarily due to its simplicity. Moreover, prompts often lack examples (25 in total), which could help guide expected reasoning or output from agents. Including examples is especially useful when agents hallucinate during agentic loops.

b) Reasoning-based prompts need a boost: One of the main features of LLM agents is their capacity for reasoning. However, reasoning cues are largely absent from prompts. This absence is due to underreporting and reliance on agentic frameworks. In any case, a study of prompt cues that improve reasoning capabilities is an open research area. In addition, many other prompt strategies remain unexplored and studies on their efficacy may yield new insights.

RQ7

Q: What are the main prompt engineering strategies followed in agents?
A: Role, instruction and zero-shot prompting.

- Studies on the use of examples may improve the quality of results.
- Reasoning prompting strategies are underreported. Further studies on their use are needed.
- Many other prompt engineering strategies, such as learning-based, iteration-based, and generative-based strategies remain unexplored. Investigating their use may uncover novel strategies.

8) RQ8: More than half of the studies (23 in total) propose agentic RSs that have at least one integration with another system. The most common integration is with a database of various types. We summarize findings in the following paragraphs.

a) Traditional RSs and databases are still options: Studies that integrate with other RSs for recommendation generation tend to rely on traditional RSs. The main reason is that most studies propose a full agentic RS when other components are agentic. Regarding databases, relational databases, especially SQL-based ones, are common. They often store item or user information. NoSQL, vector[17], and other types of databases (tuples) are alternative ways of storing information.

b) Less common integrations can give insights: Studies also integrate with the internet for item searches, knowledge graphs, Langchain, maps and Wikipedia. In all integrations, we see a need for additional information, especially about users and items. Investigating agent or system-driven dynamically

[17]https://github.com/facebookresearch/faiss

TABLE XVIII
USE CASES OF AN AGENTIC RECOMMENDER SYSTEM

Use Cases	Description
Direct Recommendation	Suggests items based on a user's past behavior.
Sequential Recommendation	Suggests items considering the order of user actions.
Chat	Engages in a dialogue with the user.
Question Answering	Answers questions about the items.
Preference Elicitation	Gathers user preferences explicitly or implicitly.
Rating Prediction	Attempts to estimate ratings for an item.
Recommendation Explanation	Justifies why an item is recommended.
Feedback	Users respond to improve future recommendations.
Review Summarization	Condenses reviews into informative summaries.
Domain-Specific	Processes data according to the needs of a specific field.

updated databases of item information remains an open research area. Moreover, improvements in service identification, interoperability, and response structure remain open research problems.

RQ8

Q: What integrations are agentic recommender systems reported to have?
A: Traditional RSs and relational databases are somewhat common.

- NoSQL, vector, and external knowledge bases are other options for information storage.
- Langchain and similar frameworks remain limited, though promising.
- Given many integration opportunities, service identification, interoperability, and structured response formats remain open research questions.

9) RQ9: Agentic RSs have uses beyond simple recommendations. We summarize use cases of ARSs identified in the studies in Table XVIII and analyze the results in the following paragraphs.

a) Direct and conversational recommendations are trending: Studies (18 in total) explore the capabilities of LLMs in chatting and intelligent reasoning to provide recommendations to users through conversation and based on a list of features. Sequential recommendations are based on the order or timing of user interactions and provide an additional layer of information that can be further explored.

b) Important use cases need to be investigated: User feedback can tailor recommendations even further while also making RSs more realistic. However, feedback is not considered in the majority of studies. Domain-specific recommendations consider information specific to the domain in recommendations, such as beats per minute (BPM) in music or focal lenses in movies, and represent a research area that remains unexplored.

c) Many open research areas: Iteration-based recommendations, where preferences are gathered in a conversation with the agent, is trending. However, there is a lack of datasets for this purpose, which hinders research in this area. Studying how timed features can be supported, rather than summarized as historical interactions often are, enables richer analysis. Lastly, process-based representations of each use case of agentic RSs, including the agents, enable another level of analysis closer to software engineering, such as the identification of artifacts or the analysis of provenance.

RQ9

Q: What use cases are agentic recommender systems reported to have?
A: Table XVIII summarizes 10 uses.

- Direct and conversational recommendations are common, exploring LLM capabilities.
- Sequential recommendations are dependent on the model's ability to understand order or time.
- There is a lack of datasets for conversational recommendations, hindering research.
- Process-based representations of use cases enable advanced analysis.

10) RQ10: Traditional recommendation strategies, such as content-based filtering and collaborative filtering, do not apply to LLMs because the reasoning process is unknown. For this reason, we report on the information used to generate recommendations. As expected, user and item information are the most commonly used data (19 in total) for recommendations, followed by proposals using additional information from internet searches (6 in total). Although the internet is the largest repository of unstructured information, well-suited for LLM agents, it remains underexplored for several reasons, including its complexity, high response time, and lack of trust. Nevertheless, studies should mitigate these risks, perhaps with integrations with search engines, and propose agentic RSs that retrieve additional information from the internet for better recommendations.

a) LLM agents and their own knowledge: Two proposals rely on the LLM agent's own knowledge about items (e.g., movies) to provide recommendations. Although training LLMs is a time-consuming and expensive process, real-time knowledge updates are a research area that may improve how recommendations are generated.

b) Recommendation explanations to understand LLM agent reasoning: As discussed, the recommendation strategy used by LLM agents when generating recommendations is not well-understood. Therefore, asking the agent to explain its reasoning is a useful approach. Aggregating, comparing, and reporting explanations may help identify patterns of reasoning in recommendation generation, but this remains an open question.

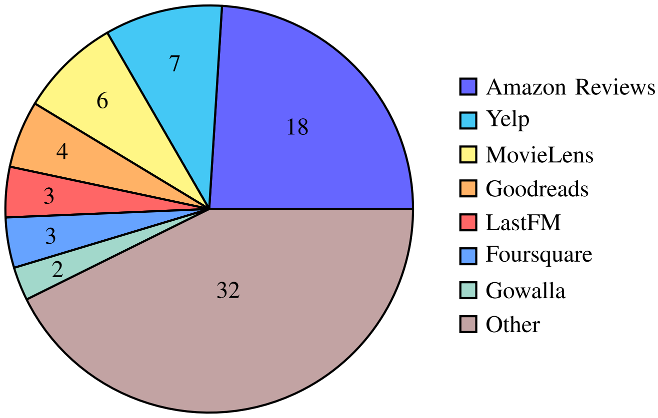

Fig. 5. A visual representation of the use of datasets in the studies analyzed.

c) Competition to improve: Adversarial strategies are not explored in the studies analyzed. These strategies may improve the quality of recommendations, potentially uncovering novel results, yet they remain an open research area.

RQ10

Q: What recommendation strategies are agentic recommender systems reported to have?
A: Strategies are mostly based on user and item information.

- Traditional recommendation strategies are no longer a major focus in current research.
- Real-time knowledge updates may enable advanced recommendations.
- Recommendation explanation patterns may help developers and researchers understand the strategies used.
- Adversarial strategies remain unexplored in the agentic recommendation domain.

11) RQ11: A substantial number of studies (36 in total) report one or more datasets used during evaluation. We have found two proprietary datasets and two studies relying on the LLM agent's knowledge. One study generated its own synthetic dataset. Underreporting is not a concern. Fig. 5 illustrates the use of datasets, followed by an analysis.

a) Amazon products and MovieLens lead in use: Amazon Products is a collection of datasets spanning categories such as books, movies, and music while MovieLens is a popular movie dataset. The reason for this result is the user familiarity with product or movie search online or in streaming services, as well as giving explicit or implicit feedback by buying an item or browsing away. However, there is a need for datasets with a broader range of items, such as food options, apps, or clothing, which are also items users frequently shop for.

b) The lack of conversational datasets is a common issue: Several conversational agentic RSs reported this issue. Mitigation strategies involve transforming lists of interactions into conversations using LLMs or leveraging transcripts of human conversations. Two excluded studies [119], [120] propose conversational datasets, which are presented here for reference. This result highlights the early stage of conversational technologies and its potential for future growth.

c) Advanced constructs remain unexplored: Other types of datasets include spatial-temporal information, real-time user interaction and feedback, and enhanced context information. However, proposals only use static user and item data to generate recommendations, leaving these research directions open.

RQ11

Q: What datasets are used in agentic recommender system evaluation and in what domains are agentic recommender systems evaluated?
A: Amazon product reviews and MovieLens are frequently used.

- A broader range of items and domains enrich recommendations, such as food or clothing, or apps.
- Synthetic datasets in data-scarce domains could be generated and used to support early stage evaluation.
- Conversational datasets are scarce, but generation or the use of transcripts could mitigate this issue.
- Advanced datasets should include spatial-temporal, real-time, or contextual information for improved recommendations.

12) RQ12: We identified numerous metrics for evaluating agentic RSs. We categorize them in four distinct groups, which are presented in Table XIX. The most common metrics are nDCG@k, HitRate@k, and Recall@k. Eight studies did not report any metrics. Among LLM metrics, ROUGE and BLEU are popular and are used for recommendation explanations. System-level metrics can be extended by incorporating software engineering knowledge. Finally, domain-specific metrics are rarely used but they appear occasionally in studies. More traditional and alternative metrics for recommender systems are found in [121].

a) Software engineering non-functional requirements are metrics: The result in Table XIX indicates efforts to evaluate agentic RSs beyond accuracy or precision. Software engineering non-functional metrics, such as reliability, performance, or security, may assist developers and researchers in evaluating agentic RSs and remain an open research avenue.

b) No standardized way of reporting experimental results: There is no clear explanation of the choice of metrics and authors sometimes confuse metrics, such as accuracy and precision. Moreover, custom metrics are not comparable, and sometimes the parameter k for the number of items in the final recommendation list is not defined. Studies are needed on more structured and standardized ways to report agentic RS evaluation.

c) Simulation frameworks assist in agentic RS evaluation: Several studies are excluded from this systematic review because of an exclusion criterion related to simulation

TABLE XIX
EVALUATION METRICS REPORTED IN THE STUDIES ANALYZED

Traditional RS metrics	nDCG@k HitRate@k Recall@k Accuracy MRR Precision@k Success rate@k F1-score Popularity@k Number of items TurnsToRecommend@k MAE MaxFreq@k Miss Rate
LLM metrics	ROUGE BLEU Fluency Coherence Conversion rate Coverage rate Entropy Explanation score Informativeness Language diversity Novelty Number of hallucinations Perspicuity Proactivity Quality of response Relational warmth Stimulation
System-level metrics	User satisfaction Acceptance rate Adherence Rate Attractiveness Dependability Efficiency Jailbreak prevention rate System usage frequency Usability Utility
Domain-specific metrics	Clinical Relevance Score Factual@k GMV (gross merchandise value) Sales-Win-Rate (custom)

TABLE XX
EVALUATION BASELINES REPORTED IN THE STUDIES ANALYZED

Baseline Type	Examples
RS	AgentCF, MACRec, MACRS, SASRec
Algorithm	BM25, MF, Popularity, random
LLM	GPT-4, Llama 2
Reasoning	ReAct, CoT, Plan&Solve, Reflexion

frameworks. However, they are listed here for reference, as they may help to improve agentic RS evaluation. This represents a growing research area. The studies are [122], [123], [124], [125], [126], [127], [128].

RQ12

Q: What metrics are used in agentic recommender systems evaluation?

A: nDCG@k, HitRate@k, and Recall@k are common. Numerous alternative metrics are shown in Table XIX

- Four groups of metrics capture different aspects of agentic RSs.
- Software engineering non-functional requirements could serve as a basis for system metrics.
- There is a need for structured and standardized ways of reporting agentic RS evaluation.
- Simulation frameworks could assist in evaluation. Although excluded from this review, they are presented in the text for reference.

13) RQ13: Studies report 107 different baselines used to evaluate agentic RSs. On average, studies that reported at least one baseline included about five, demonstrating strong efforts to evaluate their proposals. Sixteen studies did not report any baseline, which diminishes their impact and raises concerns about underreporting. We classify baselines in four groups, which are shown in Table XX.

a) RSs and LLMs are standard baselines; alternative reasoning strategies need incentive: As expected, the majority of studies use either traditional RSs or other agentic RSs as baselines in their evaluations. LLMs are also used as baselines, leveraging their embedded knowledge. This trend is expected to grow. Few studies vary their agents' reasoning strategies during evaluation, even though reasoning is a central part of the proposal. This is an important aspect that authors should address.

b) Complex execution requirements hinder research: Authors often download and execute other agentic RSs for evaluation studies. This process can be time-consuming and highly complex, especially when documentation is not helpful. Approaches and frameworks for simpler ways to download and execute agentic RSs, especially for evaluation, are in high demand.

c) An LLM fine-tuned for recommendations: The use of generic LLMs as baselines for recommendation tasks is unfair as these LLMs often tend to perform poorly. An LLM fine-tuned for recommendation tasks keeps comparisons fairer, and the development of frameworks for fine-tuning them based on specific datasets is an interesting research avenue.

d) Evaluation frameworks are in need: Many studies compare their proposals with the same baseline, such as AgentCF [54], matrix factorization, or ChatGPT. Frameworks to assist evaluation studies can help developers and researchers automate this process, yet remain an unexplored area of research.

TABLE XXI
METRIC VALUES FOR nDCG@k AND RECALL@k REPORTED ON THE STUDIES ANALYZED

		Direct		Sequential	
		nDCG@k	Recall@k	nDCG@k	Recall@k
AgentCF [54]	@1	0.2333	-	0.2133	-
DCICDRec [51]	@3	0.0503	0.0452	-	-
i2Agent [84]		-	-	0.5767	-
ARAG [82]	@5	0.4394	-	-	-
AgentCF [54]		0.4373	-	0.4379	-
RecMind [58]		0.0935	-	0.0396	-
DCICDRec [51]		0.0703	0.0742	-	-
InteRecAgent [69]		-	0.6505	-	-
RAH! [52]	@10	0.5524	0.5339	-	-
ChatCRS [63]		0.5490	-	-	-
AgentCF [54]		0.5405	-	0.5198	-
RuleAgent [86]		0.2083	0.0942	-	-
RecMind [58]		0.1607	-	0.0480	-
TAIRA [71]		-	-	0.7314	-
Zoom [85]		-	-	0.0415	-
InteRecAgent [69]	@20	0.6386	-	-	-
ChatCRS [63]		0.5530	-	-	-
RuleAgent [86]		0.1942	0.1511	-	-
LIKR [64]		-	-	0.2380	0.0946
LIKR [64]		-	-	0.1916	0.0483
Zoom [85]		-	-	0.0460	-

RQ13

Q: What baselines are used in agentic recommender systems evaluation?
A: RSs and LLMs.

- Evaluation of alternative reasoning strategies in proposals is becoming increasingly common.
- Complex deployment processes and lack of documentation hinder comparison with some systems in evaluation studies.
- Evaluations against LLMs fine-tuned for recommendations provide fairer comparisons than generic models.
- Automated execution and evaluation against popular systems, LLMs, and algorithms are limited by the lack of supporting frameworks.

14) RQ14: We select two popular and important metrics from Table XIX in RQ12, namely nDCG@k [129] and Recall@k [130], and analyze their values in the proposed agentic recommender systems. The results are presented in Table XXI. We provide a brief review of the metrics, followed by an analysis of their values.

Normalized discounted cumulative gain (nDCG) [129] measures how well the ranking of results matches the ideal ranking, by considering both relevance and item position, rewarding relevant items that appear early in the list. It is calculated from two components, the discounted cumulative gain (DCG) and the ideal DCG (IDCG), as shown in Equation 1, producing values between 0 and 1, with 1 indicating a perfect ranking.

$$\begin{aligned} \text{DCG@}k &= \sum_{i=1}^{k} \frac{2^{rel_i} - 1}{\log_2(i+1)} \\ \text{IDCG@}k &= \text{DCG of ideal ranking} \\ \text{nDCG@}k &= \frac{\text{DCG@}k}{\text{IDCG@}k} \end{aligned} \tag{1}$$

- rel_i: relevance score of the item at position i.
- i position in the ranked list (starting from 1).
- $\log_2(i+1)$: discounts lower-ranked items.

Recall [130] is a metric used in statistical analysis to measure the completeness of results. It evaluates whether the results include the expected items. Its formula is straightforward and is shown in Equation 2. Recall values range from 0 to 1, but are usually multiplied by 100 to be expressed in percentages. High recall values are closer to 100%, but must be balanced with precision. The trade-off between precision and recall is often managed using the F1-score [131].

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

- True Positives (TP): Correctly predicted relevant items.
- False Negatives (FN): Relevant items the system missed.

a) The performances of ARAG, AgentCF, i^{2}Agent, InteRecAgent, and RAH! are highlighted: From the table, we notice that ARAG [82] and AgentCF [54] achieve good nDCG values at low k, in direct recommendations. ARAG uses agentic RAG as one of its main features. This result is reported on a clothing dataset. As for sequential recommendations, i^{2}Agent [84] achieves a higher nDCG score than AgentCF with a lower k. i^{2}Agent's dynamic memory mechanism and improvements from individual feedback contribute to this result on an Amazon Movies dataset. InteRecAgent [69] has a good recall value for a k value of 5, but no nDCG@k measure reported. A good balance is found in RAH! [52] for direct recommendation at $k = 10$. This result is attributed to RAH!'s attempt to model user personality and its novel Learn-Act-Critic agentic loop.

b) Results vary slightly with evaluation choices: The choice of dataset, LLM model used, number of agents, and other evaluation parameters influence the values of nDCG@k, Recall@k, and other metrics. We present the best results, as reported by the studies, using the recommended system configuration and excluding ablation studies. For detail analysis, please refer to the original studies.

c) Report nDCG@k and Recall@K: At a minimum, studies are encouraged to report both nDCG@k and Recall@k metrics for comparison with other agentic RSs. These are important metrics that evaluate agentic RSs in two different dimensions: quality and completeness of results. HitRate@k is also a popular and simple metric to report. Development processes, including testing tasks, encourage developers to adapt to evaluation practices.

d) Testing frameworks for agentic systems are needed: Evaluating any agentic system can be a challenging task given the many configurations that must be tested. Ablation studies mitigate this problem, but they remain time-consuming. Frameworks that support the evaluation, comparison, and reporting of metrics are lacking and would greatly assist developers and researchers. This remains an open research area.

e) Modularization and individual testing assist researchers in explaining results: Currently, a low nDCG or Recall value often remains unexplained because of the many components involved in an agentic RS. Frameworks that help developers modularize the system and test components individually can clarify results and address performance issues more efficiently.

RQ14

Q: What nDCG and Recall values for agentic recommender system evaluation are reported?
A: Table XXI shows nDCG@k and Recall@k values for direct and sequential recommendations.

- ARAG's agentic RAG and i^{2}Agent's feedback mechanism improve results, while RAH!'s Learn-Act-Critic agentic loop enhances recall.
- Results depend on configuration, and we report the best cases.
- Frameworks for evaluation, comparison, and reporting, as well as for system modularization and component testing assist researchers in evaluation and in explaining results. This remains an open research area.

15) RQ15: About half of the studies (21 in total) report research gaps, open research questions, or directions for future work. We also analyze the issues addressed in the papers, which represent research problems. We classify research gaps and other gaps found in the studies analyzed in four groups: software engineering core, software engineering application, agentic AI, and psychology. Table XXII summarizes the findings. An detailed analysis follows.

TABLE XXII
RESEARCH GAPS IN AGENTIC RECOMMENDER SYSTEMS

Software Engineering Core
• Lack of architectural patterns
• Lack of contracts for agentic workflows
• Lack of testing frameworks
• The observability problem
• Memory management
• Privacy
• Distributed learning
• Dynamic choice of planning or reasoning strategies
• Metadata and provenance support
• Real-time processing
• Malicious prompts

Software Engineering Applications
• Knowledge graph integration
• Data scarcity
• Summarization of long user histories
• Personalization vs. Privacy
• Multi-step reasoning plans and decisions
• Cost-aware planning and orchestration
• Asynchronous pipelines
• Resource-efficient recommendation and personalization

Artificial Intelligence
• Underspecified recommendation goals
• Multi-objective recommendations
• Intent modeling
• Customization of agents for personalized recommendations
• Cross-domain preferences
• Generalization under sparse data
• The popularity bias problem
• Variety-optimized agent strategies
• Hallucinations

Psychology
• Anthropomorphism
• Lack of psychological taxonomies
• Persuasion
• Psychological factors in memory modeling

We begin with open research problems in the **software engineering core** and agentic RSs.

a) Lack of architectural patterns: Agentic recommender systems are typically composed of multiple agents and components. However developers and researchers lack guidelines on architectural principles for arranging LLM agents and components. This problem becomes more concerning given the growing number of tools that LLM agents are expected to handle. Future research should establish architectural patterns by:

(i) prototyping agent layouts inspired by established paradigms such as Model-View-Controller (MVC);
(ii) evaluating microservices-based designs for their impact on performance, maintainability, and adaptability;
(iii) systematically comparing existing novel and architectural patterns in terms of how they accommodate agent memory, shared memory, and tool access;

(iv) developing benchmarks and best practices to guide practitioners.

b) Lack of contracts for agentic workflows: Despite their advanced reasoning capabilities, LLM agents often struggle to produce outputs in the expected format, a challenge that becomes more acute in multi-agent systems such as agentic recommender systems. Researchers should:

(i) systematically evaluate libraries such as Pydantic 5 and protocols such as Harmony[18] and MCP[19] for their effectiveness in enforcing structured outputs across diverse agent interactions;

(ii) design communication protocols that not only transmit the next message but also encode metadata such as recommendation priorities, execution strategies, intermediate steps, and generation times in programmatic formats;

(iii) develop benchmarks to compare prompt-embedded versus protocol-based approaches will clarify trade-offs in reliability, scalability, and interoperability, ultimately guiding best practices for structured communication in agentic recommender systems.

c) Lack of testing frameworks: During the development of agentic RSs, there is limited support for testing agentic constructs, such as memory, tool chaining, or multi-turn plans. Frameworks for unit or integration testing of these constructs assist developers to deploy improved systems. Future research should:

(i) design standardized testing frameworks for agentic constructs;

(ii) develop benchmarks to evaluate memory, tool chaining, and multi-turn planning;

(iii) compare unit versus integration testing approaches in agentic RSs;

(iv) investigate automated methods for detecting failures in complex agent interactions.

d) The observability problem: Researchers would like to identify what parts of an agentic RS are responsible for producing high-quality or low-quality recommendation lists. In more serious cases, regulatory audits require this information to be available. However, this remains a challenge because of the difficulties in tracing agent decisions and results to system components, such as prompts, knowledge sources, or memory items. Ablation studies mitigate this problem, but more direct ways are required. Future research should:

(i) develop methods to trace recommendation quality back to specific system components;

(ii) design mechanisms to support regulatory audits by making agent decisions transparent;

(iii) improve tracing techniques beyond ablation studies to capture the role of prompts, knowledge sources, and memory items more directly.

e) Memory management: Efficient methods of retrieving and updating the user memory by LLM agents are still under investigation. Studies often involve large amounts of prompt-related data, although only a relevant subset is typically required. Approaches to memory pruning are being explored, but systematic solutions remain limited. Advancing this area requires:

(i) developing efficient retrieval mechanisms that identify and access only relevant subsets of user memory;

(ii) designing update strategies that maintain accuracy while minimizing redundancy;

(iii) evaluating pruning techniques for balancing memory size with recommendation quality;

(iv) establishing benchmarks to compare different memory management approaches in agentic recommender systems.

f) Privacy: Training or fine-tuning agents should be conducted with caution because of the risks of including personally identifiable information (PII). Sometimes the agent's responses may become unpredictable increasing the likelihood of data leakage. Addressing these challenges requires:

(i) developing methods, such as differential privacy or data anonymization techniques, to prevent PII from being incorporated during training or fine-tuning;

(ii) designing mechanisms to detect and mitigate unpredictable outputs that risk exposing sensitive data;

(iii) evaluating protocols for secure handling of agent responses in multi-agent or user-facing systems;

(iv) establishing auditing practices to ensure compliance with privacy regulations.

g) Distributed learning: Distributed learning, particularly federated learning, necessitates protocols that preserve user privacy and result integrity. Advancing this area requires:

(i) designing privacy-preserving protocols that prevent exposure of sensitive user data during distributed training;

(ii) developing mechanisms that ensure the integrity and consistency of results across federated learning systems.

h) Dynamic choice of planning or reasoning strategies: Ablation studies are often conducted to examine reasoning strategy selection. A framework for dynamic choice helps developers enable agents to adopt the most suitable strategy. Such a framework can also enable the reuse of plans and performance histories for each strategy. Future research should:

(i) construct a framework that supports dynamic selection among reasoning strategies;

(ii) enable systematic reuse of plans and performance histories to improve efficiency;

(iii) develop evaluation methods to measure the effectiveness of dynamic strategy choice in agentic recommender systems.

i) Metadata and provenance support: The formats, storage, and management of metadata and provenance in agent communication are critical elements that must remain queryable and securely maintained. This information includes message origins, consent records, and processing steps. The lack of frameworks to support these activities indicates that this is an open research problem. Actionable future research should:

(i) construct frameworks to query and securely maintain metadata and provenance;

[18]https://github.com/openai/harmony
[19]https://modelcontextprotocol.io/

(ii) design methods to record and trace message origins, consent records, and processing steps;
(iii) evaluate how such mechanisms improve auditability and trust in agentic recommender systems.

j) Real-time processing: Unlike static movie reviews, spatial-temporal information challenges agentic recommendation systems by requiring real-time POI recommendations. The research area remains open as studies have not addressed this problem. More specifically, studies should address real-time changes to agentic architectures and service availability, adaptations of agent plans in response to new information or cost constraints, and the stream processing of data in the recommendation process. Advancing this area calls for:

(i) designing scalable real-time architectures that maintain responsiveness under dynamic workloads;
(ii) developing mechanisms to guarantee continuous service availability in multi-agent environments;
(iii) creating adaptive strategies that allow agents to adjust plans in real time without degrading performance.

k) Malicious prompts: Only one study has examined agents in the context of security. Open problems in this emerging area of research include input sanitization, intent classification, ethical considerations, and sandboxing. Future research should:

(i) develop systematic methods for input sanitization to prevent malicious or corrupted data from influencing agent behavior;
(ii) design robust intent classification techniques to detect and handle adversarial or ambiguous inputs;
(iii) investigate frameworks that embed ethical considerations into agent decision-making;
(iv) construct secure sandboxing environments to isolate agent actions and reduce security risks.

Open research problems in **software engineering applications** and agentic recommender systems include the following.

a) Knowledge graph integration: Knowledge graphs can significantly augment the capabilities of LLM agents, supported by a rich body of literature. Efficient querying and standardized communication of results and explicit specification of limitations remain under active development. Future research should:

(i) design software frameworks that enable efficient querying of large-scale knowledge graphs, for example through optimized graph database engines;
(ii) establish standardized APIs or communication protocols to ensure consistent exchange of results across agentic systems;
(iii) implement mechanisms for explicitly encoding and communicating the limitations of knowledge graph outputs to downstream agents and users.

b) Data scarcity: Several studies note the scarcity of conversational datasets, often resorting to synthetic data generation. Transfer learning and related strategies may be explored to address this open research problem. Key directions for investigations include:

(i) investigating methods for generating high-quality synthetic conversational data;
(ii) exploring transfer learning strategies, such as fine-tuning agents on related domains (e.g., customer support or educational dialogues) before adapting them to recommender contexts;
(iii) systematically evaluating hybrid approaches that combine synthetic data generation with transfer learning to mitigate data scarcity while maintaining recommendation quality.

c) Summarization of long user histories: Some studies employ agents to summarize user history. Challenges related to efficiency and the justification of agent decisions remain open research questions. Advancing this area requires:

(i) developing scalable summarization algorithms that can handle long user histories without excessive computational overhead;
(ii) designing mechanisms for explainable summarization, where agents provide justifications or evidence for the decisions made based on condensed histories;
(iii) evaluating trade-offs between summary length, efficiency, and decision quality in agentic recommender systems.

d) Personalization vs. Privacy: The tradeoff between personalization and privacy may be mitigated by the choice of relevant strategies such as differential privacy [132], [133], [134]. However, this research area remains largely unexplored. A second tradeoff with performance is also relevant, but insufficiently investigated. Further experimental studies are required and strongly encouraged. Advancing this area calls for:

(i) implementing differential privacy mechanisms in agent training pipelines and measuring their impact on personalization quality;
(ii) conducting systematic experiments to quantify the tradeoff between privacy guarantees and system performance, for example by benchmarking latency and accuracy under varying privacy budgets;
(iii) developing evaluation frameworks that balance personalization, privacy, and performance, enabling reproducible comparisons across agentic recommender systems.

e) Multi-step reasoning plans and decisions: Current agentic RSs, whether conversation or not, initially gather item and user information and generate recommendations in a single step. However, advanced approaches engage the user iteratively to elicit feedback and to ensure that recommendations are both personalized and of high quality. Further work is needed to:

(i) construct frameworks that support iterative recommendation cycles, allowing agents to refine outputs based on user feedback;
(ii) design mechanisms for tracking and reusing intermediate reasoning steps to improve transparency and efficiency;
(iii) evaluate the impact of multi-step reasoning on personalization quality and user satisfaction in agentic recommender systems.

f) Cost-aware planning and orchestration: Instead of planning and executing each step, LLM agents can generate multiple plans, estimate their costs across performance, priority, time, or other metrics, and select the most appropriate to produce a recommendation. Further development in this area should:

(i) construct frameworks that allow agents to generate and compare alternative plans under varying cost metrics;
(ii) implement evaluation mechanisms that quantify trade-offs between performance, latency, and resource consumption;
(iii) design orchestration strategies that adaptively select plans in real time to balance efficiency and recommendation quality.

g) Asynchronous pipelines: As the number of agents grows and they become more specialized, asynchronous pipelines can address performance challenges. In addition, such agentic pipelines lead to improved resource utilization and enhanced scalability. Further exploration in this area should:

(i) design scheduling mechanisms that coordinate asynchronous agent tasks while minimizing bottlenecks;
(ii) implement monitoring tools to track resource utilization and detect inefficiencies in pipeline execution;
(iii) evaluate scalability through experimental studies that measure throughput and latency under increasing agent loads.

h) Resource-efficient recommendation and personalization: The use of the compression pattern can efficiently reduce the size of user or item profiles, or the user-item interaction matrix. The distillation pattern, when a larger model guides the learning of a smaller model, may help researchers address the tradeoff between performance and personalization. The adapter pattern can be used to enable agent communication with other systems, including other agentic RSs, for improved recommendations. Further investigation in this area should:

(i) evaluate compression techniques such as matrix factorization or embedding pruning to balance efficiency with recommendation accuracy;
(ii) implement and benchmark distillation pipelines where smaller models inherit personalization capabilities from larger teacher models;
(iii) design adapter-based integration strategies that allow agents to interoperate across heterogeneous systems while maintaining resource efficiency.

We discuss open software engineering research questions within recommender systems from an **artificial intelligence** perspective.

a) Underspecified recommendation goals: Users often lack clarity about the types of recommendations they seek. For instance, a user might write "I want to watch a movie that makes me happy", although the specific movie features that produce this mood are uncertain. Agentic RSs should incorporate mechanisms, such as algorithms, agentic strategies or data structures (e.g. graphs), to assist users in articulating their needs and to refine goals and sub-goals efficiently. Progress in this area requires:

(i) developing algorithms that can infer latent user preferences from vague or underspecified queries;
(ii) constructing agentic strategies that iteratively elicit clarifications and refine user goals through dialogue;
(iii) designing graph-based data structures that represent evolving goals and sub-goals, enabling agents to track and update user intent over time.

b) Multi-objective recommendations: Users may specify several recommendation goals, such as cost, release year, or specific features. In addition, users might require recommendations in heterogeneous formats, such as audio, video, images, or physical products. Multi-objective strategies that identify appropriate constraints and access the suitable datasets remain an open research problem. Advancing this area requires:

(i) developing algorithms that can balance multiple objectives simultaneously, for example by applying multi-objective optimization techniques to recommendation tasks;
(ii) constructing methods to encode and prioritize user-specified constraints such as cost or release year within agentic pipelines;
(iii) designing data integration strategies that allow agents to retrieve and combine heterogeneous formats (e.g., audio, video, images, products) into coherent recommendation outputs.

c) Intent modeling: User preferences evolve over time and in response to contextual factors, such as weather, mood, or situation. Moreover, short- and long-term user intentions may diverge. LLM agents should understand changes in user intent, negotiate with users, or recover from disruptions when appropriate. Negotiation prompts and experimental studies with users remain largely unexplored research directions. Future research should:

(i) develop models that capture both short-term contextual signals (e.g., mood or location) and long-term preference trajectories;
(ii) design negotiation prompts that allow agents to elicit clarifications and resolve conflicts between diverging user intentions;
(iii) conduct experimental studies with users to evaluate how agents recover from disruptions and maintain coherent intent modeling over time.

d) Customization of agents for personalized recommendations: LLM agents should have different personalities based on several factors including the user age or level of expertise, the items under consideration, or contextual information such as the weather conditions or time of the day. Customization is often specified in the prompt, but can also be encoded in training. Reusable personalities and benchmarks can help researchers achieve better results in addressing this open research question. Future research should:

(i) develop methods for encoding reusable agent personalities directly into training pipelines, enabling consistent personalization across contexts;
(ii) construct benchmarks that systematically evaluate how personality customization affects recommendation quality and user satisfaction;
(iii) design adaptive mechanisms that adjust agent personality dynamically in response to contextual signals such as time of day or weather conditions.

e) Cross-domain preferences: User preferences in one domain may correlate with user preferences in another domain, which advanced agent capabilities may capture. Mapping may rely on explicit feedback or inferred user personality traits.

The representation of cross-domain preference (e.g. user-item matrix or embedding vectors) remains a significant research challenge. Advancing this area requires:

(i) designing algorithms that learn mappings between domains using explicit feedback, such as ratings or reviews;
(ii) investigating methods for inferring personality traits or latent factors that explain correlations across domains;
(iii) developing representation techniques, for example multi-domain embedding vectors or joint user-item matrices, that capture cross-domain relationships in a scalable and interpretable way.

f) Generalization under sparse data: Most users do not explicitly communicate their preferences to the system. Few-shot prompting, personas modeling, and preference prediction may help address this open problem. Future research should:

(i) investigate few-shot prompting strategies that enable agents to generalize from minimal user input;
(ii) develop persona modeling techniques that capture implicit preferences and contextual cues when explicit data is limited;
(iii) design preference prediction algorithms that infer likely user choices under sparse data conditions, and evaluate their effectiveness across diverse recommendation scenarios.

g) The popularity bias problem: Blockbuster movies are watched and rated more frequently, which often leads to their overrepresentation in recommendations. LLM agents should generate recommendations that justify biases, while agentic loops should enable users to reject overly popular recommendations. For commercial agentic RSs, biases may be deliberately introduced for advertising purposes. Addressing this gap calls for:

(i) designing algorithms that explicitly detect and mitigate popularity bias while preserving recommendation variety;
(ii) constructing agentic loops that allow users to override or reject overly popular items and refine their preferences;
(iii) investigating transparency mechanisms that justify when popularity bias is applied, especially in commercial contexts where advertising incentives may influence recommendations.

h) Variety-optimized agent strategies: LLM agents need to balance exploitation and exploration through specific prompt instructions. Encouraging exploration may yield unexpected results without diminishing user trust. Advancing this area requires:

(i) designing prompt strategies that explicitly control the trade-off between exploitation of known preferences and exploration of novel items;
(ii) developing evaluation frameworks that measure how exploration impacts user satisfaction, trust, and long-term engagement;
(iii) constructing adaptive mechanisms that dynamically adjust exploration levels based on contextual signals or user feedback.

i) Hallucinations: LLM agents can hallucinate. Adversarial approaches that validate outputs can mitigate this problem. Advanced approaches include algorithms for consistency checks and scoring mechanisms to prevent meaningless recommendations. Conversational checks and the incorporation of user feedback are additional strategies. Future research should:

(i) develop adversarial validation techniques that systematically detect and filter hallucinated outputs in recommendation pipelines;
(ii) design algorithms for consistency checks and scoring mechanisms that quantify the reliability of generated recommendations;
(iii) explore conversational strategies and user feedback loops that enable agents to recover from hallucinations and maintain user trust.

We finally describe the **psychological** aspects of software engineering research gaps in agentic recommender systems.

a) Anthropomorphism: Incorporating human-like characteristics into LLM agents can enhance user connection and trust. Assigning an agent a name or appearance, adapting its language to user mood, and tailoring its behavior to contextual factors such as time of the day represent strategies that can be explored to increase user engagement, improve feedback and reviews, and enhance recommendation quality. Future research should:

(i) investigate how different forms of anthropomorphism, such as names, visual appearance, or conversational style, influence user trust and engagement;
(ii) design adaptive mechanisms that allow agents to adjust language and behavior dynamically in response to user mood or contextual signals;
(iii) construct benchmarks and experimental studies that measure the impact of anthropomorphic features on recommendation quality, feedback, and long-term user satisfaction.

b) Lack of psychological taxonomies: No studies of user behavior currently guide LLM agents in sustaining conversations with users. Users may feel secure, anxious, or avoidant, and LLMs should account for these states when generating recommendations. The absence of systematic measurements and empirical tests keeps this research question open. Future research should:

(i) develop psychological taxonomies that classify user states such as security, anxiety, or avoidance in the context of recommendation interactions;
(ii) design measurement instruments and systematic protocols to empirically evaluate how these states influence user engagement and trust;
(iii) construct agent strategies that adapt recommendations dynamically based on psychological taxonomies, ensuring sensitivity to user well-being and conversational context.

c) Persuasion: Similar to human sellers, persuasion strategies can be encoded in prompts to guide LLM agents in recommending items aligned with organizational business goals. Item metadata can assist LLM agents in identifying products suitable for advertisement. Future research should:

(i) investigate how different persuasion strategies, such as framing, social proof, or scarcity cues, can be embedded into prompts without undermining user autonomy;
(ii) design methods that leverage item metadata to personalize persuasive recommendations while maintaining transparency;
(iii) conduct empirical studies to evaluate the ethical and commercial impacts of persuasive agent strategies, including user trust, satisfaction, and long-term engagement.

d) Psychological factors in memory modeling: Currently, LLM agents maintain a memory of user interactions and conversations. Psychological characteristics should be incorporated into memory, as they reflect deep values that guide user preferences. Several studies have begun to investigate the inclusion of personality traits in recommendation processes, and this research area is expected to expand. Future research should:
(i) develop frameworks for integrating psychological characteristics, such as personality traits or motivational states, into agent memory representations;
(ii) design methods to evaluate how psychological factors influence long-term recommendation quality and user satisfaction;
(iii) conduct empirical studies that test adaptive memory models capable of updating user profiles in response to evolving psychological signals.

RQ15

Q: What research gaps are reported in the studies?
A: We classify open research areas in four groups and list them in Table XXII.

- Quality-related features of agentic RSs remain underexplored.
- Advances in communication protocols, memory, and multi-domain, multi-goal, and multi-interaction settings remain open research areas.
- Personalization, agent customization, and hallucination reduction may be enhanced through prompt design or model training.
- Anthropomorphism, psychological taxonomies of agent behavior, and memory adaptations incorporating psychological traits may improve personalization, increase user engagement and yield better recommendations.

IV. Threats to Validity

This systematic literature review aims to characterize the novel field of agentic recommender systems. However, many factors may impact the steps towards this goal, such as the novelty of the field, the scarcity of results, and the understanding of the new concepts. In this section, we discuss threats to the validity of this study and how we mitigate them.

a) Internal Validity: We included arXiv[20] preprints in this study. Although some of the preprints are published studies, many are not published, and thus, not peer-reviewed. This raises questions about the quality of some of the studies in this review. We created robust exclusion criteria to reject low-quality studies and analyze only studies with a good level of development. In addition, authors can decide the evaluation parts that are used to assess the proposals, such as datasets, metrics, and model parameters. The choice of these parts impacts the results and thus the conclusions in this study. New systematic studies should be performed to reinforce our conclusions. Lastly, the studies we inspect are fixed at a single point in time. However, these studies evolve, as new ideas emerge. Our attempts to find the most recent version of the proposed agentic recommender systems mitigate this threat.

b) External Validity: Advancements in the field of LLMs are as recent as 2022. As a result, publications in the fields of agentic AI and agentic RSs are usually not older than 2023. We retrieved studies for this systematic review in September 2025, and thus included studies published in 2023, 2024 and part of 2025. The constrained number of years is seen as a threat as including more years can strengthen the conclusions. For the same reasons, we only inspected 44 studies. The relatively low number is seen as a threat to validity since studies should be a representative set of the field. To mitigate both threats, we increased the number of search engines and include arXiv preprints. In addition, the studies included in this systematic review are mostly academic and not operating in a business setting, where costs, privacy and legal constraints are taken into consideration. We expect the conclusions of this review to be technical and encourage new studies to investigate the impacts of these business concerns. Lastly, the field of AI or ML changes rapidly and some of the conclusions in this study may need to be updated. To mitigate this, regular systematic reviews on agentic recommender systems are encouraged to check what conclusions still hold.

c) Construct Validity: Inspecting the studies and retrieving some pieces of information to answer the research questions can be a challenging process because these pieces may not be explicitly stated or they may be available in the reported repository. We see this as a threat to validity despite the efforts to retrieve accurate information. In addition, since the goal of this systematic review is to characterize agentic recommender systems, some concepts are not well-defined, which can impact our ability to understand the studies. For example, the distinction between the Manager and Recommender agents was identified late in the review. We revise our understanding before stating the conclusions. Lastly, this systematic review aims to be broad and to cover many perspectives around agentic recommender systems. However, some of these perspectives can be missing. We then encourage communication with the corresponding author and new characterization studies to make sure that the field is well-characterized and researchers and practitioners have access to high-quality information and conclusions.

[20]https://arxiv.org

V. Conclusion

Agentic recommender systems (ARSs) base their recommendation workflow on the collaboration between LLM agents and tools and are characterized by the use of these agents in the generation of recommendations. This research field is a result of the integration between LLMs and recommender systems (RSs) and, because of its originality, the definition and characterization of the field is still evolving. This systematic review of the literature aims at addressing these concerns by selecting, inspecting, and analyzing agentic recommender systems proposed in the literature from many different perspectives.

For this systematic literature review, our research query retrieved 44 high-quality studies, from five different sources, describing single or multiple-LLM agent recommender systems. We then examine each study with respect to its agents, agent relationships, prompts, integrations, use cases, recommendations, evaluation, and research gaps. Results are discussed across 15 main research questions, covering multiple perspectives.

Some results are worth pointing out such as a list of 13 LLM agent roles presented in Table XII, the widespread use of GPT models, shown in Fig. 1, and a framework for agentic recommender systems, which can be seen in Fig. 2. This systematic literature review has placed a strong focus on the characterization of prompts, including a generalized recommendation prompt, presented in Fig. 3 and the description of the use of many prompt items, such as the use of personas, cues, the output format, and examples, or their absence, as in zero-shot prompting (Table XVII). Other results include integrations with databases (RQ8), the popularity of direct (non-sequential) conversation recommendations (RQ9), the use of Amazon reviews, Yelp, and MovieLens as datasets, shown in Fig. 5 and the popularity of nDCG@k [129] and Recall@k [130] metrics during the evaluation of agentic recommender systems, whose values are presented in Table XXI. Lastly, numerous research gaps are discussed under RQ15 in the areas of software engineering, artificial intelligence, and psychological considerations, including contracts for agents in workflows, scarce conversational datasets, improved intent modeling, and anthropomorphism of agents.

We expect that this systematic literature review provides an understanding of the current and near future developments in this novel field of agentic recommender systems. We present results from many different perspectives for improved coverage and hope that these insights are useful for both early-career and experienced researchers in the field. For future work, we plan to update the results with more studies on new technologies, strategies, or tools related to agents, prompts, or evaluation. This is important since development in AI-related fields is shifting rapidly.

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